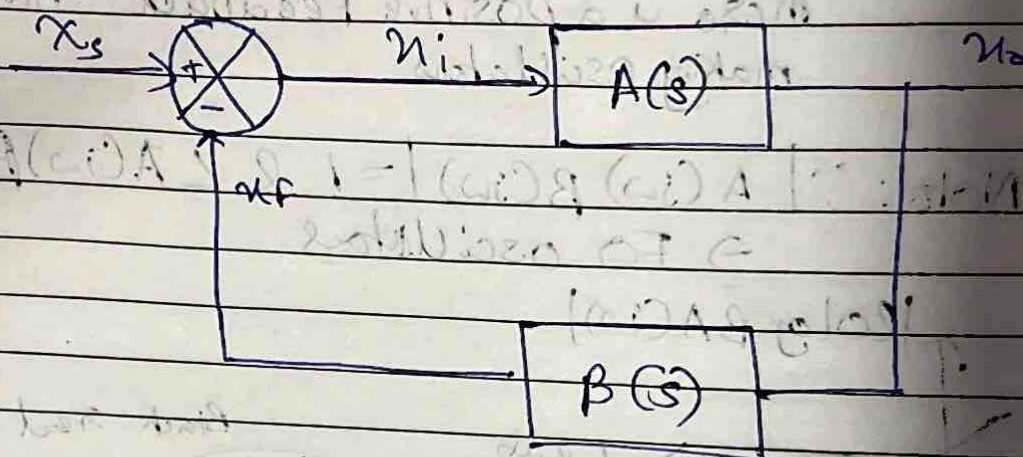


23/08/24

classmate

Date _____
Page _____# Stability: $x_i e^{s_i t}$ $C_{\omega 180^\circ}$ 

$$A_F(s) = \frac{A(s)}{1 + A(s)B(s)}$$

$$A_F(j\omega) = \frac{A(j\omega)}{1 + A(j\omega)B(j\omega)} \quad s = \sigma + j\omega$$

$$A(j\omega) \cdot B(j\omega) = |A(j\omega)B(j\omega)| e^{j\phi(\omega)}$$

→ at $\phi(\omega) = 180^\circ$

$$A(j\omega) \cdot B(j\omega) = -|A(j\omega)B(j\omega)|$$

Note: If $A \cdot B = -1 \rightarrow$ Gain = ∞

& $n_f = -n_i$ i.e. n_i is oscillating in the ckt at freq of ω_{180°

→ ~~$A(j\omega)B(j\omega) = -1$~~

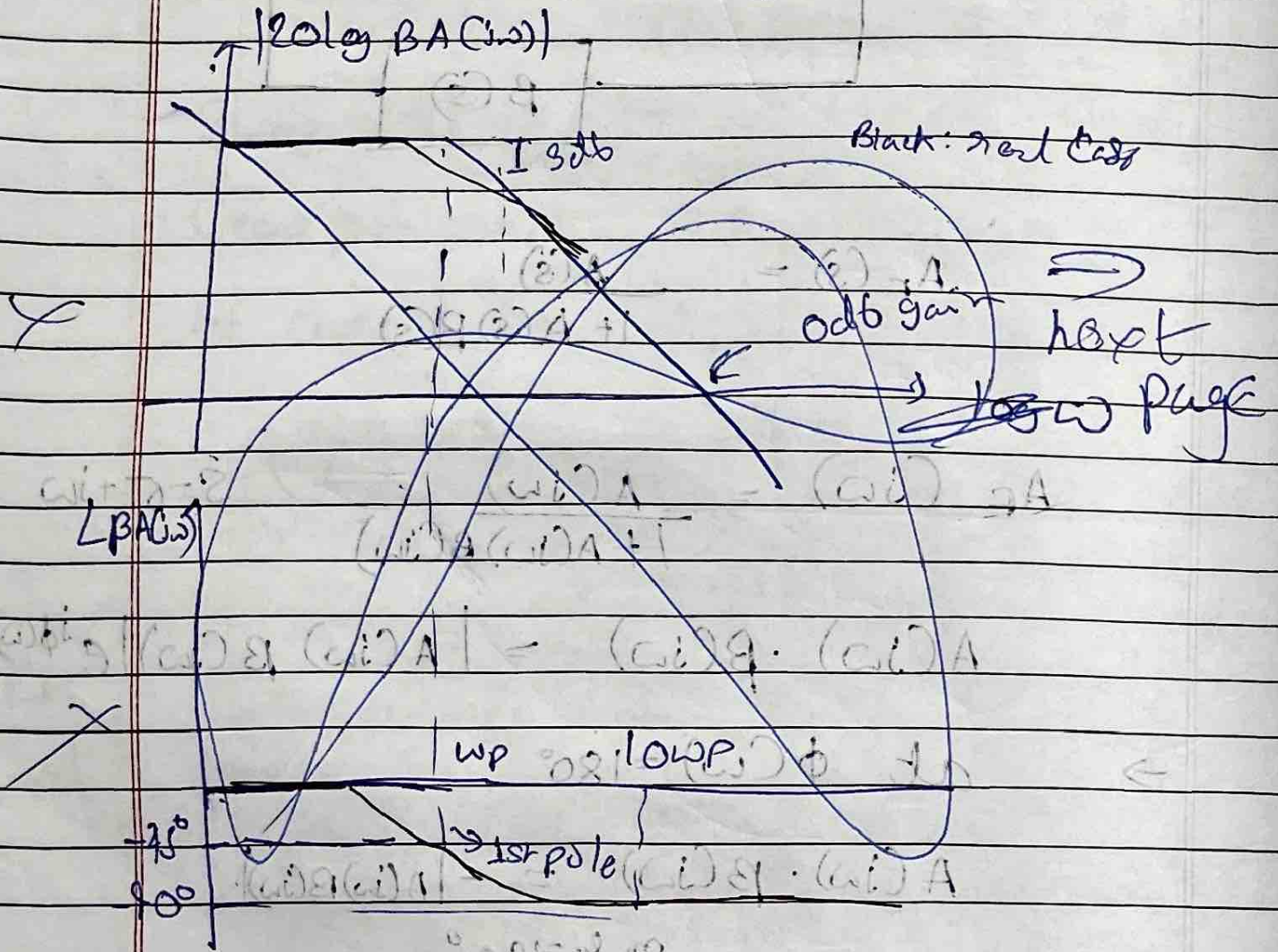
In decade $\Rightarrow$ Ratio is const

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Note: When $A_{mag} > 1$ & $\phi = 180^\circ$
there is a positive feedback : ~~used for~~
making oscillators

Note: $\therefore |A(j\omega)B(j\omega)| = 1$ & $\angle A(j\omega)B(j\omega) = 180^\circ$
 $\Rightarrow$ For oscillations



Note: $1 + j\omega b \Rightarrow$ If $\omega \uparrow$ neglect 1

$$\angle BA(j\omega) = \tan^{-1} \frac{b\omega}{a}$$

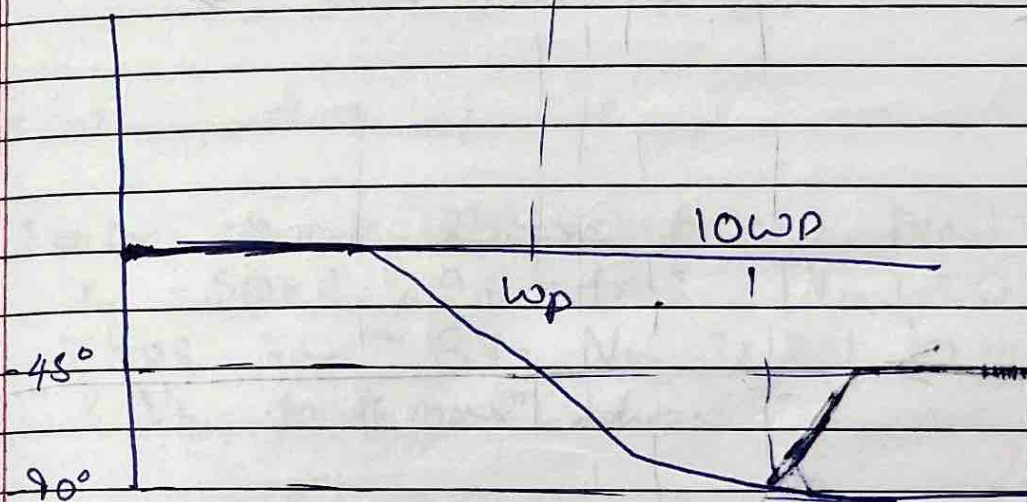
When ωb is small $\angle \approx 0$
if $\omega b \gg a \Rightarrow 90^\circ$

1 pole sys

mag

Mag drops little before ω_p (Resonance)

3dB



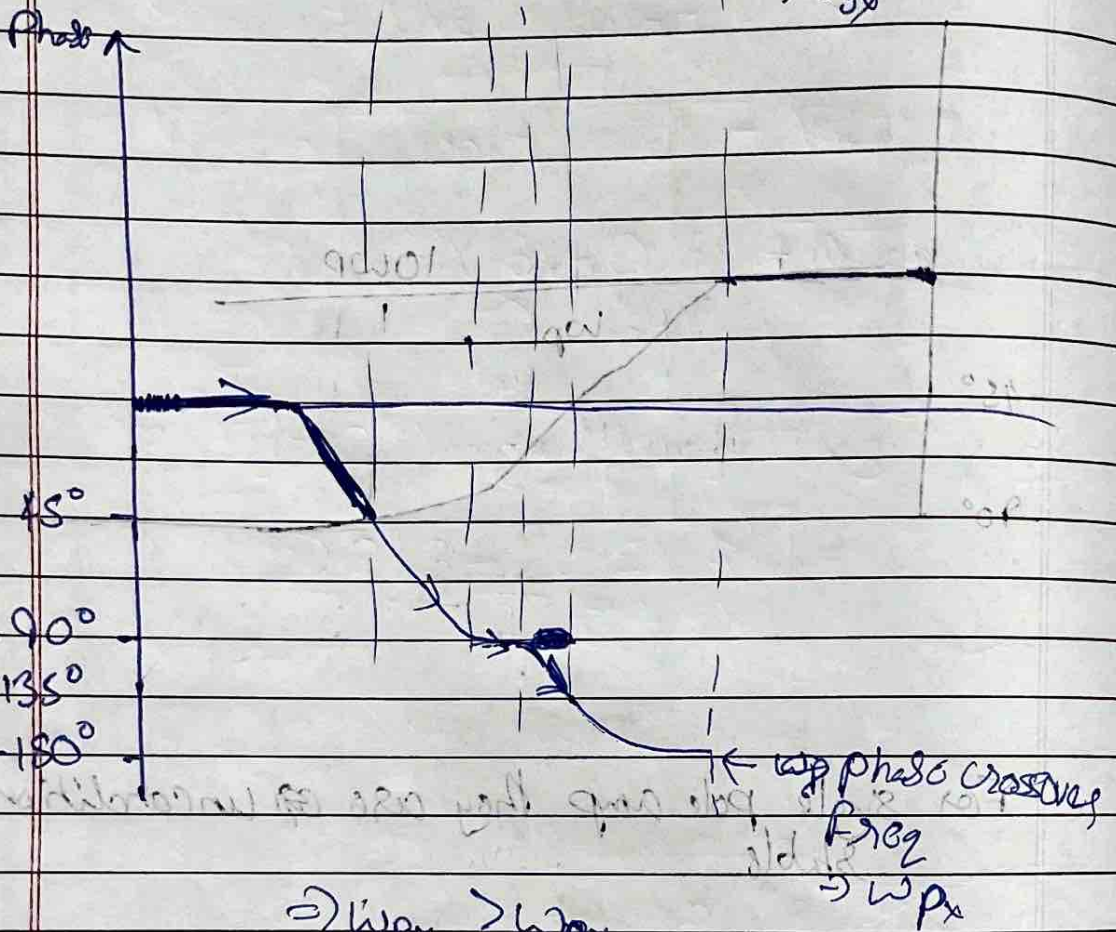
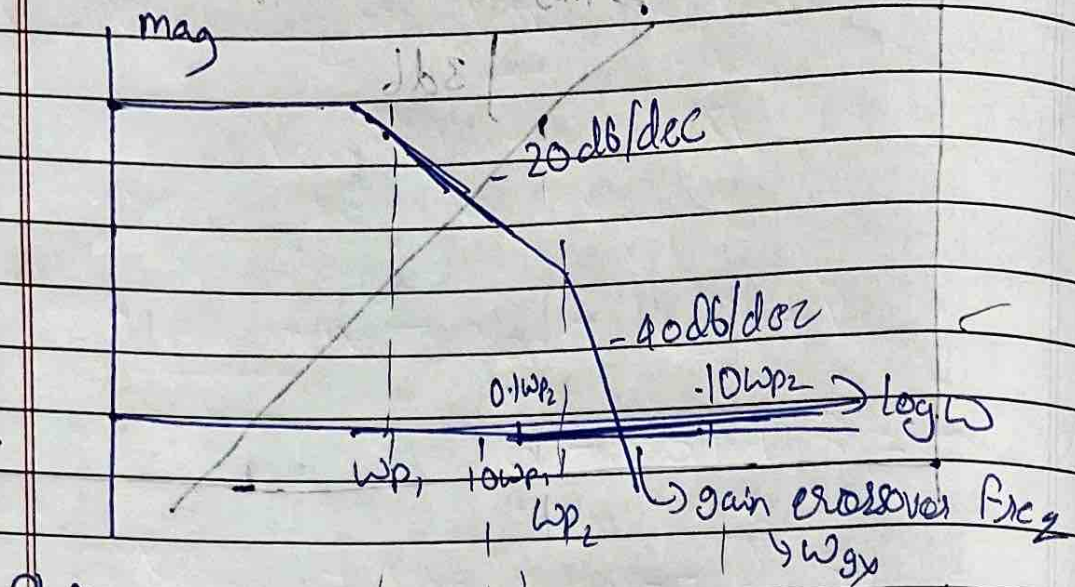
For single pole amp they are ~~un~~ unconditionally stable

Imp Ckt

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For 2 pole ckt :



$$\Rightarrow \omega_{px} > \omega_{gx}$$

$\therefore \text{phase} > -180^\circ \rightarrow \text{Stable sys}$

problem $\rightarrow \text{phase} \approx 180^\circ$ & $\text{mag} = 1$

If $\omega_{gx} = \omega_{px}$ then its a unstable sys

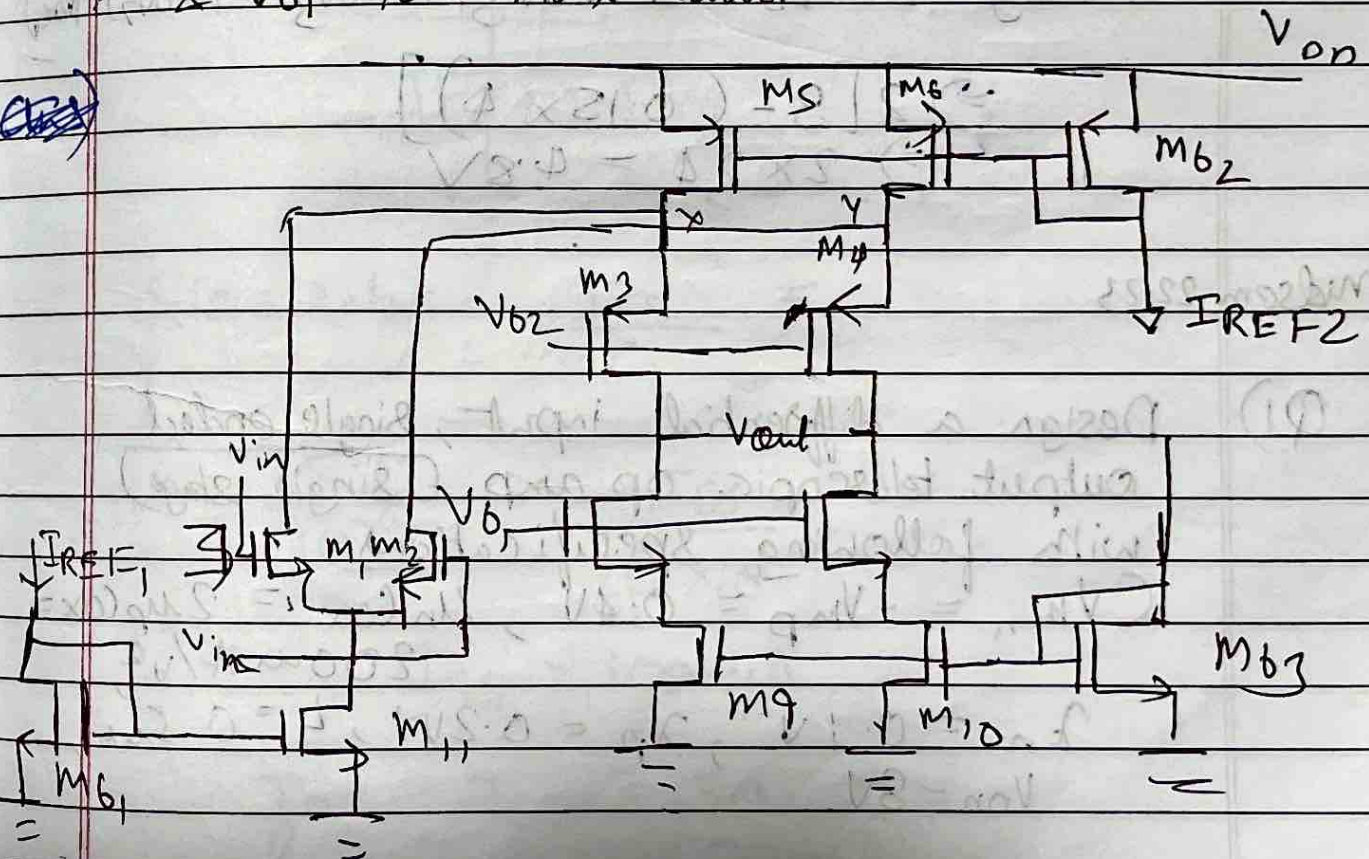
If $\omega_{gx} > \omega_{px}$ $\Rightarrow$ Again unstable sys

Phase margin: $\phi_{\omega_{gx}} > -180^\circ$ (ex: -170°)
 $\text{phase margin} = 180 - |\phi_{gx}|$

Note: If Gain margin: At ω_{px} by how much loop gain is greater than 0dB

Quiz Q5 2023

Q). For the circuit shown below, $V_{ov1} = 0.15V$, $r_o = 50k\Omega$, $g_m = 1mS$, $|V_{th}| = 0.3V$ of all X_{ox} . $V_{DD} = 3V$. V_{b2} is set to max^m value & V_{b1} to its max^m value.



ans) $V_{in\ min} = |V_{ov1}| + |V_{GS1}| = 0.15 + 0.45 = 0.6V$

$V_{GS1} = V_{ov1} + V_{th} = 0.45V$
 same for all xes

$V_{in\ max} = V_{DD} - |V_{ov5}| + V_{th}$

$= 3 - 0.15 + 0.3 = 3.15V$

$V_{b1\ min} = V_{ov1} + V_{GS1} = 0.15 + 0.45 = 0.6V$

$V_{b2\ max} = V_{DD} - |V_{ov5}| - |V_{GS1}|$

$\Rightarrow 3V - 0.15V - 0.45V$

$\Rightarrow 2.4V$

O/P Swing $= 2[V_{DD} - (|V_{ov5}| + |V_{ov3}| + |V_{ov7}| + |V_{ov1}|)]$

$\Rightarrow 2[3 - (0.15 \times 4)]$

$\Rightarrow 2 \times 2.4 = 4.8V$

Midsem 22-23

Q1) Design a differential input, single ended output telescopic op amp (single stage) with following specifications

$V_{thn} = -V_{thp} = 0.4V$, $\mu_n C_{ox} = 2\mu_p C_{ox} = 200\mu A/V^2$

$\lambda_n = 0.1V^{-1}$, $\lambda_p = 0.2V^{-1}$, $L = 0.5\mu m$, $V_{DD} = 3V$

→ For a load capacitor of 20pF , the slew rate is $5\text{V}/\mu\text{s}$

→ Output voltage swing = 1.6V

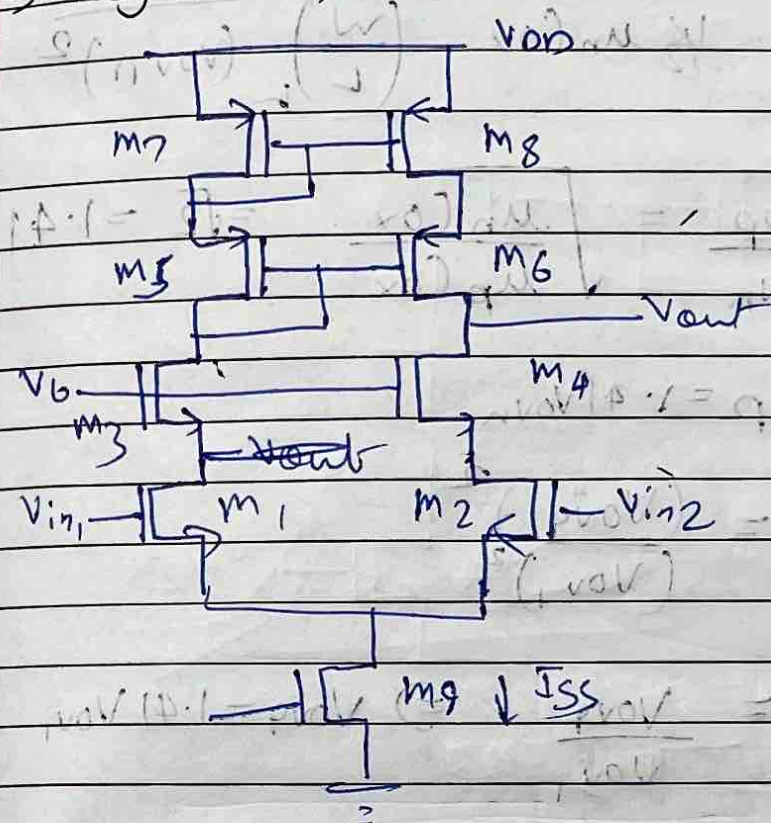
→ All xhgs are almost same size

→ Tail current source is single Xh NMOS

• Current mirrors C_{MOS}

→ neglect (π)

and



$$\text{Slew rate} = \frac{I_{SS}}{C_L} = 5 \text{ V}/\mu\text{s}$$

$$\frac{5\text{V}}{10^{-6}\text{s}} = \frac{V_{DD} I_{SS}}{20 \times 10^{-12}}$$

$$I_{SS} = 100 \mu\text{A}$$

$$\therefore I_{D1} = I_{D2} = 50 \mu\text{A}$$

$$I_{D9} = 100 \mu\text{A}$$

Single ended

$$V_{out\ swing} = V_{DD} - [V_{ov8} + V_{ov6} + |V_{thp}| - V_{ov4} - V_{ov2} - V_{ovq}]$$

$$\Rightarrow V_{ov2} + V_{ov4} + V_{ov6} + V_{ov8} + V_{ovq} = 1V$$

$$\frac{I_{Dpmos}}{I_{Dnmos}} = \frac{\frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L}\right)_p (V_{ovp})^2}{\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_n (V_{ovn})^2}$$

$$\therefore \frac{V_{ovp}}{V_{ovn}} = \sqrt{\frac{\mu_n C_{ox}}{\mu_p C_{ox}}} = \sqrt{2} = 1.41$$

$$V_{ovp} = 1.41 V_{ovn}$$

$$\frac{I_{Dq}}{I_{D1}} = \frac{(V_{ovq})^2}{(V_{ov1})^2}$$

$$\sqrt{2} = \frac{V_{ovq}}{V_{ov1}} \Rightarrow V_{ovq} = 1.41 V_{ov1}$$

$$V_{ovn} + V_{ovn} + 1.41 V_{ovn} + 1.41 V_{ovn} + 1.41 V_{ovn} = 1$$

$$V_{ovn} = 0.16V$$

$$V_{ov4} = 0.16V, V_{ov8} = 0.23V, V_{ovq} = 0.23V$$

$$50 \times 10^{-6} \quad 200 \times 10^{-6}$$

classmate

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$$I_{D1} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 \times V_{ov1}^2$$

$$\left(\frac{W}{L} \right) = 19.53 = 11.9 \left(\frac{W}{L} \right)_{5.8} = 18.9$$

$$\left(\frac{W}{L} \right)_q = 18.9$$

$$V_b = V_{ovq} + V_{ov1} + V_{ov3} + V_{thn}$$

$$\Rightarrow 0.23 + 0.16 + 0.16 + 0.4 = 0.95V$$

$$V_{in,cm,min} = V_{ovq} + V_{ov1} + V_{thn} = 0.23 + 0.16 + 0.4 \Rightarrow 0.79V$$

$$\begin{aligned} V_{in,cm,max} &= V_{DD} - V_{GS7} - V_{GS5} + V_{thn} \\ &= 3 - (0.23 + 0.4) - (0.23 + 0.4) \\ &= 2.14V \end{aligned}$$

$$Gain = -g_{m1} (g_{m6} r_{o6} r_{o8} || g_{m4} r_{o4} r_{o2})$$

$V_{swing} \approx$ (After shorting V_{in} & V_{out})

$$V_{out,max} = V_b - \{V_{GS4} + V_{th2}\}$$

$$V_{out,min} = V_b - V_{th4}$$

A. 1.26

Phase margin:

$$A_f(j\omega) = \frac{A(j\omega)}{1 + B(j\omega)A(j\omega)}$$

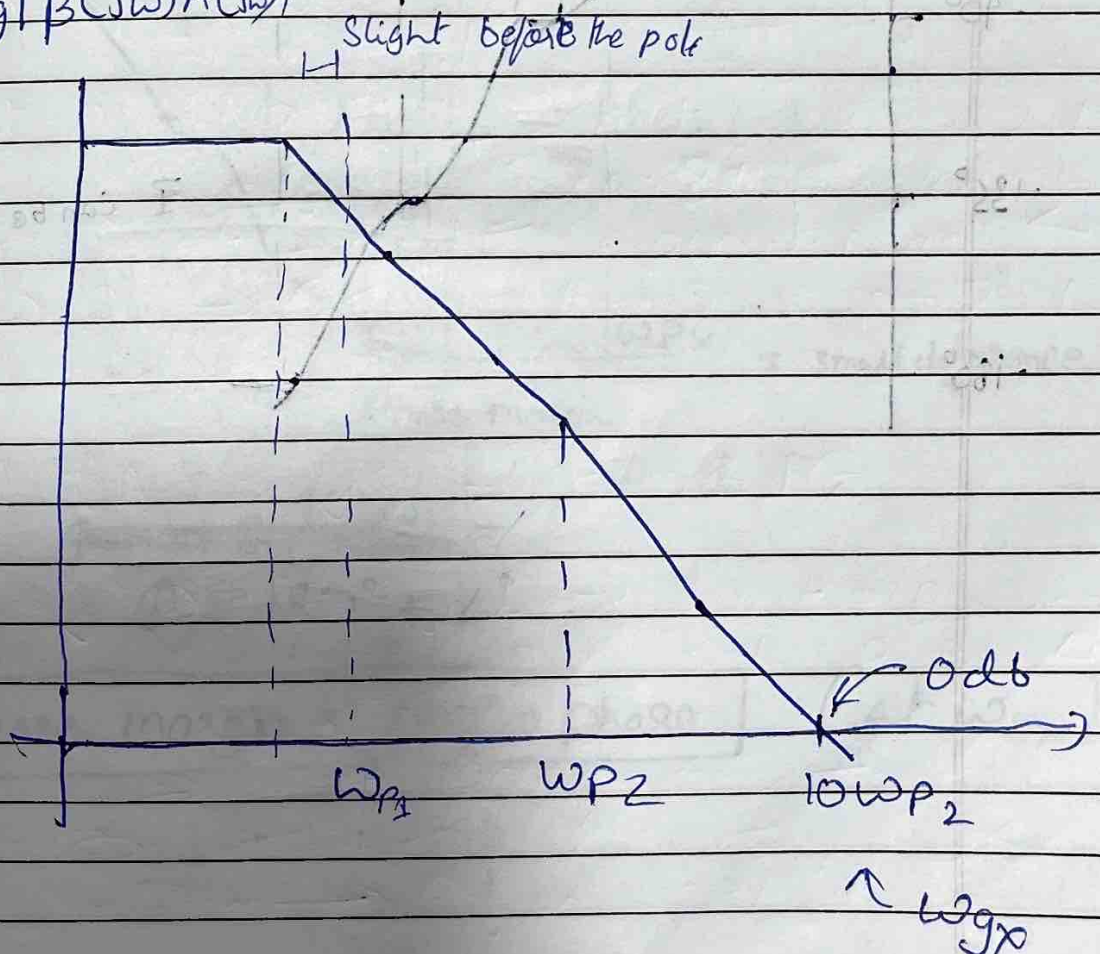
Note: If $|B(j\omega)A(j\omega)| < 1$ & $\phi < 0$

$$\boxed{A_f > A(j\omega)}$$

→ Frequency at which $|B(j\omega)A(j\omega)| = 1$ is called Oscillating Frequency

ie → $\boxed{\begin{aligned} |A(j\omega)B(j\omega)| &= 1 \\ \angle A(j\omega)B(j\omega) &= 180^\circ \end{aligned}}$ → Freq for this

$20 \log |B(j\omega)A(j\omega)|$



$a + j\omega$
 at pole: $a = \omega \Rightarrow$

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$L.B (j\omega) A(j\omega)$

$(\omega i) A = (\omega i) A$

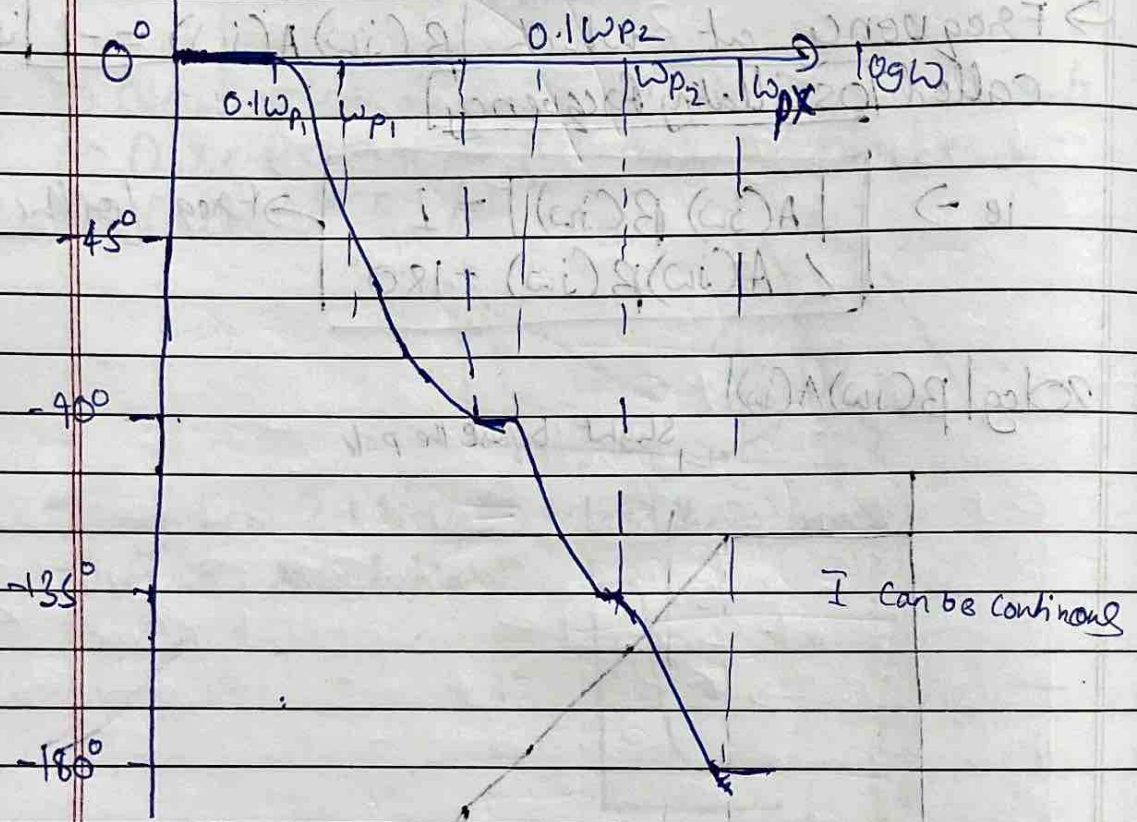
$(\omega i) A (\omega i) B$

~~scribbled out text~~

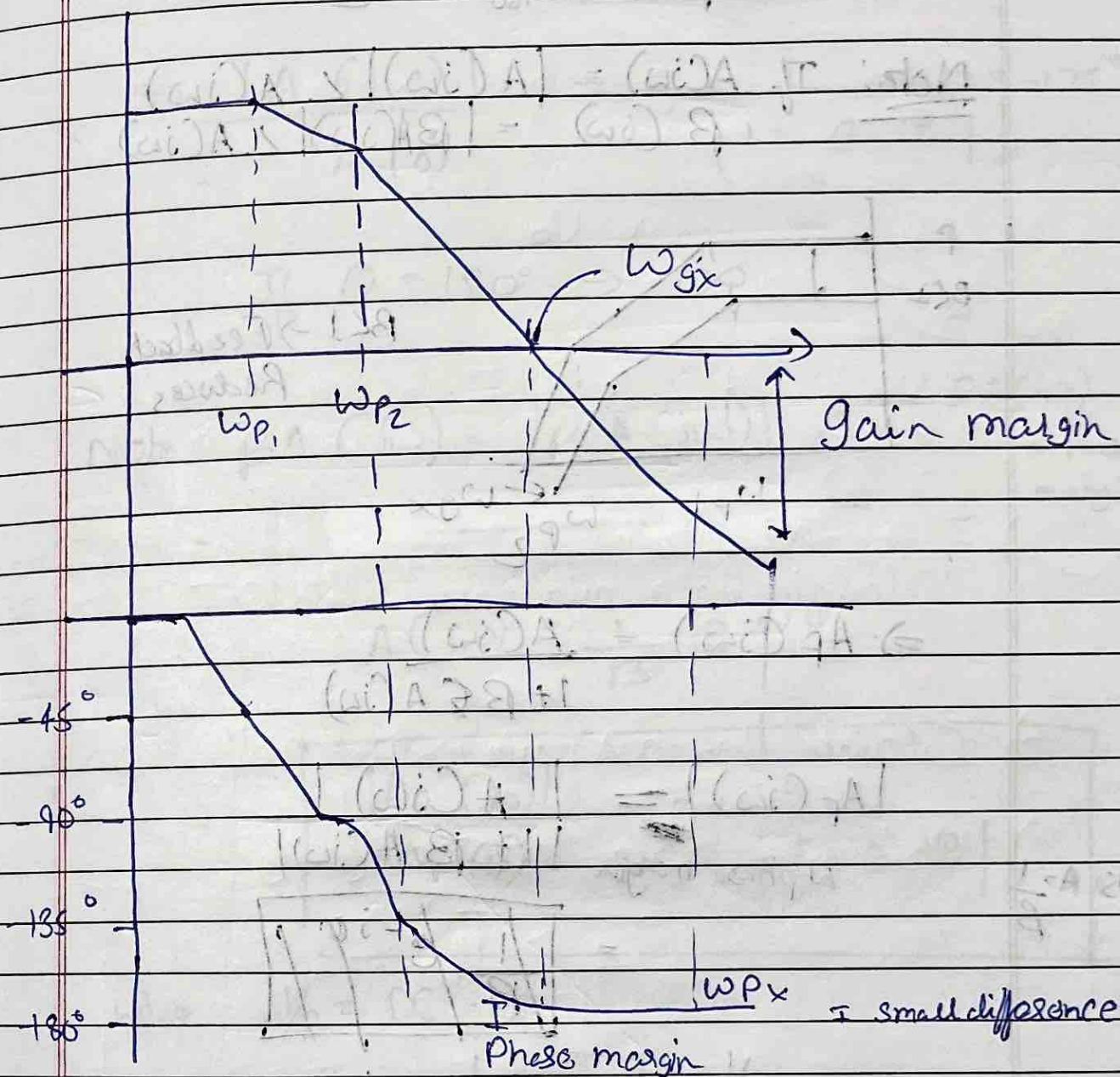
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$0 > 0 > 1 (j\omega) A (j\omega) B$

$(j\omega) A < A$



For stability $\Rightarrow \boxed{\omega_{gx} < \omega_{px}}$



$$\phi = 180^\circ + \angle$$

$$\phi = 180^\circ + \angle$$

$$\boxed{\text{Phase margin} = 180^\circ + \text{phase}} \quad (\text{at } \omega_{gx})$$

If Phase margin = 5°

Then $|B A(j\omega)| = 1$

$\angle B A(j\omega) = -175^\circ$

classmate

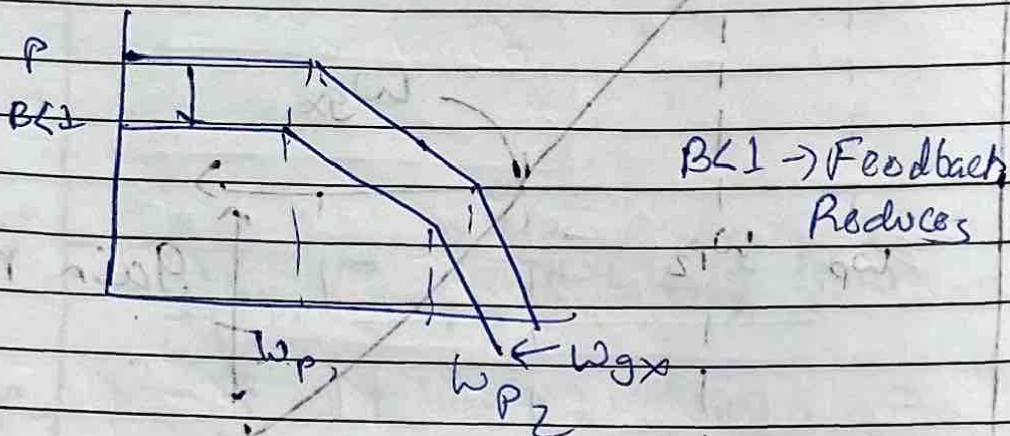
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~~$A(j\omega)$~~

$$|A(j\omega) B(j\omega)|_{180^\circ} = 1$$

Note: If $A(j\omega) = |A(j\omega)| \angle A(j\omega)$
 $B(j\omega) = |B(j\omega)| \angle B(j\omega)$



$$\Rightarrow A_f(j\omega) = \frac{A(j\omega)}{1 + B A(j\omega)}$$

$$|A_f(j\omega)| = \frac{|A(j\omega)|}{|1 + B A(j\omega)|}$$

at phase margin

Use $A = \frac{1}{B}$

$$= \frac{1/B}{1 + 1 \cdot e^{-j5^\circ}}$$

$$\text{If } A = 1/B \Rightarrow \frac{1/B}{1 + 1 \cdot e^{-j5^\circ}}$$

$$= \frac{1/B}{1 + 1 \cdot e^{-j5^\circ}}$$

$$\Rightarrow \frac{A(j\omega)}{A(0)} = \frac{1}{|1 + e^{-j0}|}$$

$\uparrow$
DC gain

$$\Rightarrow \frac{|A(j\omega)|}{A(0)} = \frac{1}{|1 + e^{-j0}|} = -175^\circ$$

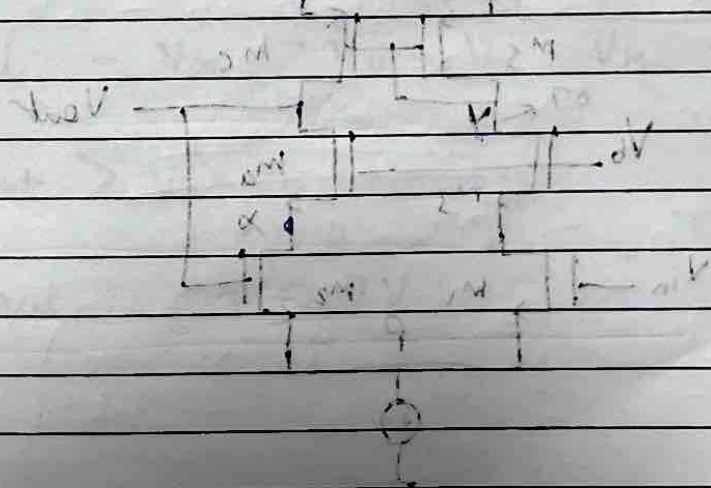
$$\text{If } 0 = 180^\circ \Rightarrow \infty$$

Note: $\angle PA(j\omega) = \angle BA(j\omega) + \angle PA(j\omega)$ at phase margin

$$A(j\omega) = \frac{1}{B} e^{-j0}$$

$$\frac{|A_c(j\omega)|}{|A_c(0)|} = \frac{1}{|1 + e^{-j0}|}$$

where $\frac{1}{B} = \text{DC gain}$



2/09/24

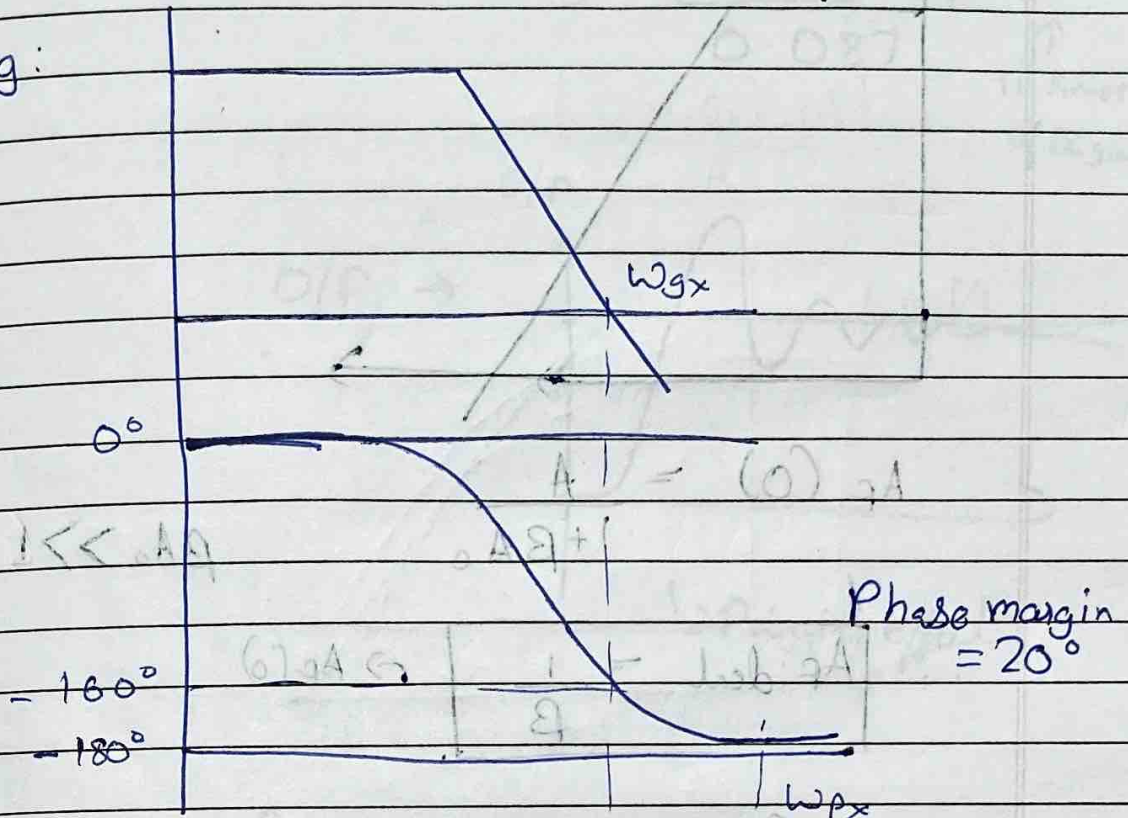
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Phasemargin cont

eg:



→ We know $1: |BA(j\omega)| \angle e^{j0} = \beta A(j\omega)$

$|0| = 180^\circ$ - phase margin at $\omega = \omega_{gx}$

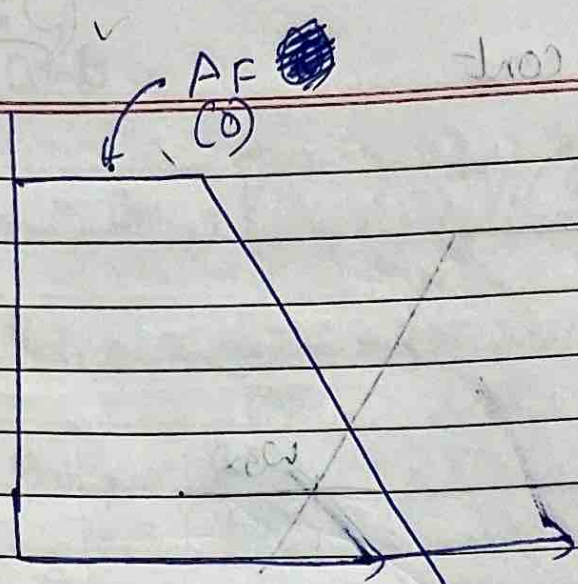
Note: $\beta A(j\omega_{gx}) = \frac{1}{e^{-j0}}$ at $\omega = \omega_{gx}$

OR $A(j\omega_{gx}) = \frac{1}{\beta} e^{-j0}$

$\therefore A_p(j\omega_{gx}) = \frac{A(j\omega_{gx})}{1 + \beta A(j\omega_{gx})} = \frac{1}{\beta} \frac{e^{-j0}}{1 + 1 \cdot e^{-j0}}$

$|A_p(j\omega_{gx})| = \frac{1/\beta}{1 + 1}$

(1)



$$A_F(0) = \frac{A}{1 + \beta A_0}$$

$$\beta A_0 \gg 1$$

$$A_{Fideal} = \frac{1}{\beta} \Rightarrow A_F(0)$$

Sub in ①

$$\frac{|A_F(j\omega_{gc})|}{|A_F(0)|} = \frac{1}{|1 + e^{-j\theta}|}$$

Case 1: When phase margin = 5°

$$\theta = 175$$

$$\frac{|A_F(j\omega_{gc})|}{|A_F(0)|} = \frac{1}{|1 + e^{-j175}|}$$

$\Rightarrow$

$$|1 + 0.996 - j0.087|$$

$\Rightarrow$

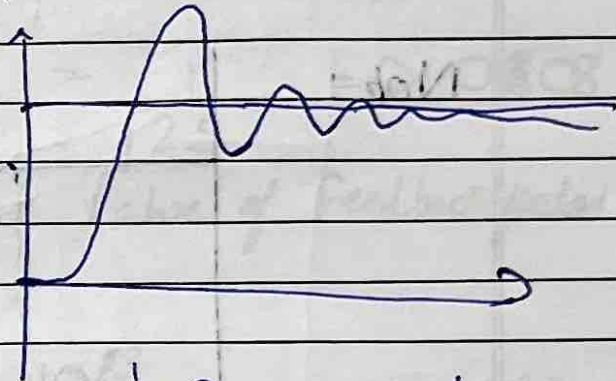
$$|1 - 0.996 - j0.087|$$

$\Rightarrow$

$$0.087$$

11 times
of α gainO/P $\rightarrow$

O/P



Pinging effect

For 45° pm

$$\frac{|A_f|}{A(0)}$$

$$= \frac{0.1}{0.121 + 1 + \cos 90^\circ - j \sin 90^\circ}$$

$$(1) \Rightarrow \cos 180^\circ = -1$$

$$1 - 1/\sqrt{2} - j 1/\sqrt{2}$$

$$\cos 180^\circ = -1$$

$$\Rightarrow \frac{1}{1 - 1/\sqrt{2} - j 1/\sqrt{2}} = 0.54129$$

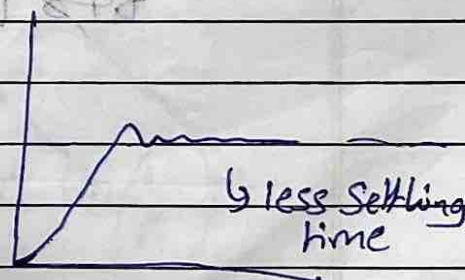
$$\approx 0.77$$

$$(100) \text{ and } 101 = 0.0184$$

$$0.02 \text{ bar } 101 \times 86 \rightarrow$$

$$\text{new to } 251 = (2 \text{ bar})$$

$$\text{ring } 100$$

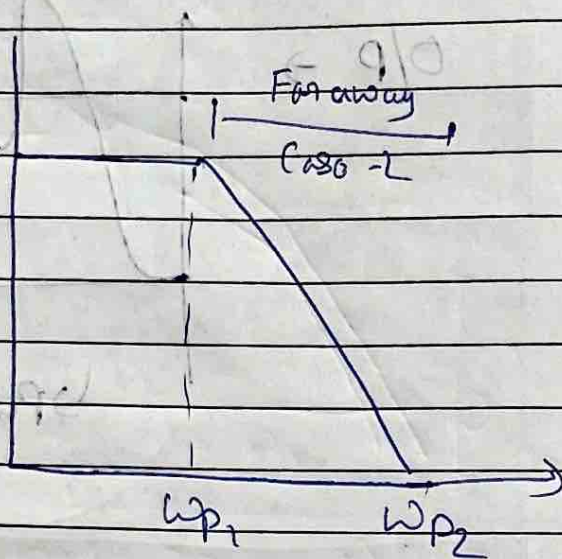
less settling
time

P.M

At $\phi = 60^\circ$

$$\left| \frac{A_f(j\omega_{gx})}{A_o} \right| = \frac{1}{1 + e^{-j120}} \Rightarrow \underline{\underline{1}}$$

Note:



$$\omega_{p2} = \omega_{gx} = \omega_{px}$$

Eg:

$$A(s) = \left(\frac{10}{1 + s/10} \right)^3$$

$$\phi = 3 \tan^{-1} \left(\frac{\omega}{10^4} \right)$$

$$180^\circ = 3 \tan^{-1} \frac{\omega}{10^4}$$

$$\omega_{180} = 10^4 \tan(60^\circ)$$

$$\Rightarrow \sqrt{3} \times 10^4 \text{ rad/sec}$$

$$A(\omega) = 125 \text{ at } \omega = 180^\circ$$

60L gain

$$\text{Loop gain} = \beta A(s) = \beta \cdot 125 \angle 1$$

$$\text{at } \omega_{180^\circ} \Rightarrow \beta < \frac{1}{125}$$

$$\beta_{\text{critical}} = \frac{1}{125} = 0.008$$

↳ critical value of feedback factor

$$\text{CL gain} = \frac{10^3}{1 + \beta_{\text{cr}} \times 10^3} \quad A(s) = 10^3$$

$$\Rightarrow \frac{10^3}{1 + 0.008 \times 10^3} \Rightarrow 111.11$$

⇒ Frequency compensation: Any desired loop gain if we can change position of poles.

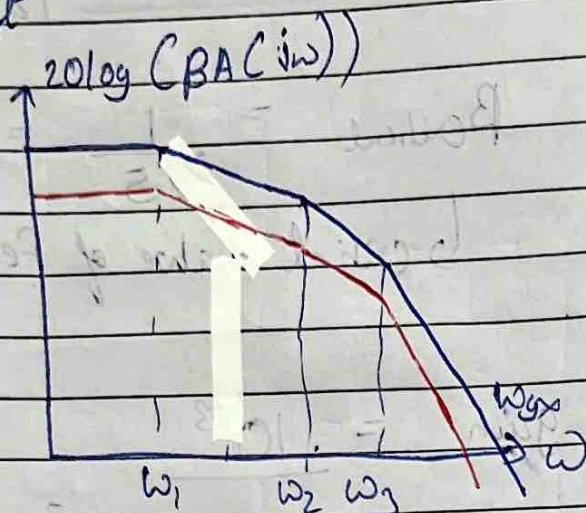
$$20 \log AB = 20 \log A + 20 \log(1/\beta)$$

DC gain is $1/\beta$ in CL

min CL gain is $1/\beta$ or $20 \log(1/\beta)$

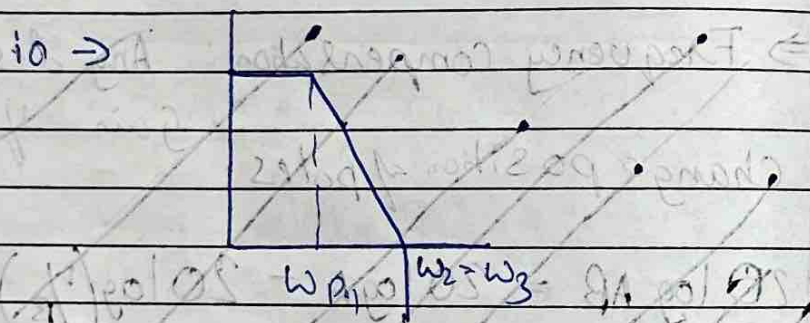
Let's assume $\omega_1, \omega_2, \omega_3, \omega_4$ are far apart

→ If we reduce $B \rightarrow \omega_{gx}$ shifts towards the left

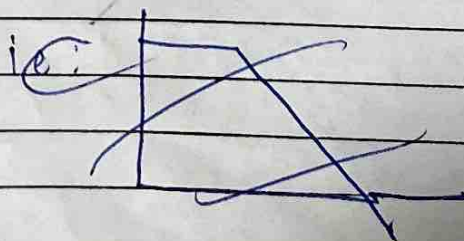


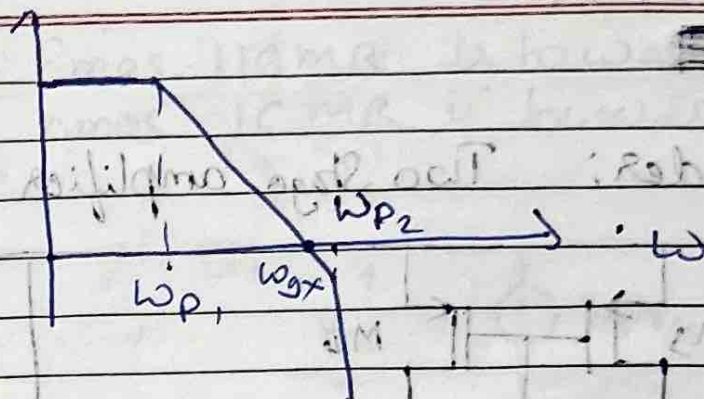
$$20 \log(CBA) = 20 \log(A) - 20 \log\left(\frac{1}{B}\right)$$

Phase margin $= 45^\circ$ when $\omega_{gx} = \omega_2$

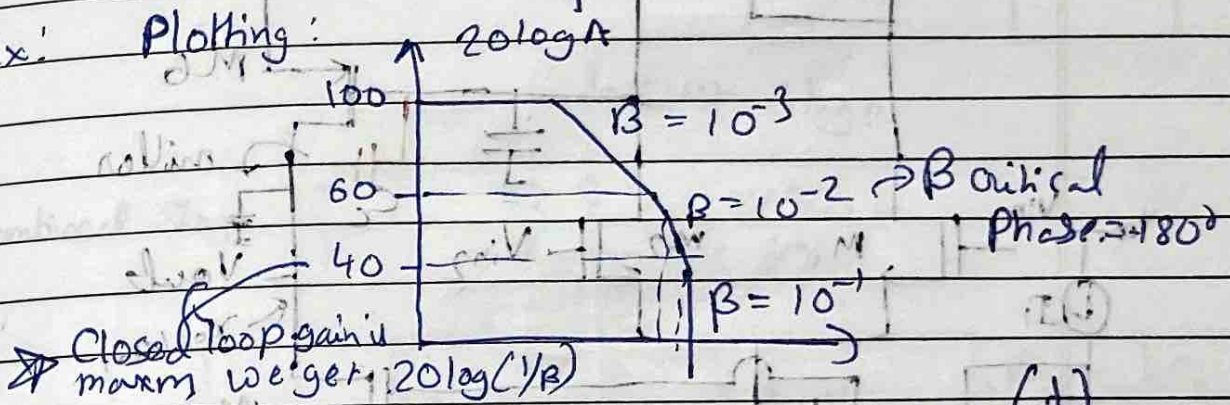


→ For the phase margin to be greater than 45° , the ω_{gx} should lie on -20 dB/decade line. The first pole should be before ω_{gx} & second pole should be after ω_{gx} .



$20 \log (AB)$ 

Ex: Plotting:

As we increase β $20 \log \left(\frac{1}{\beta} \right) \downarrow$

At point of intersection

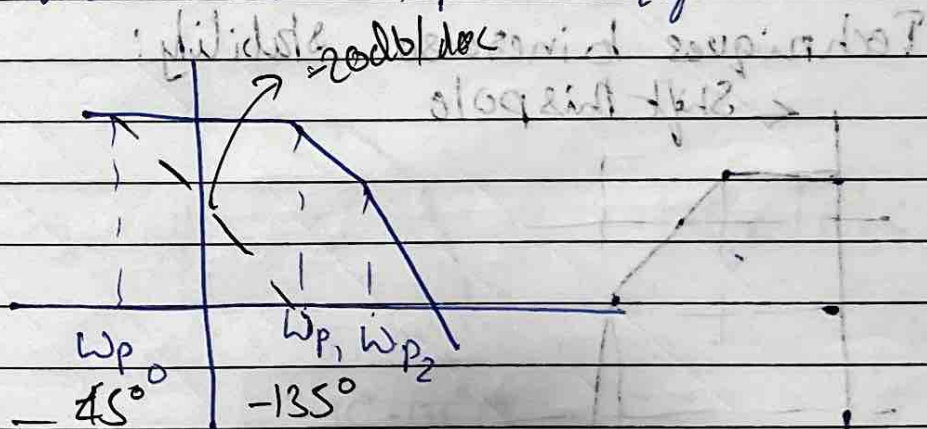
leaning towards unstable

$$20 \log (AB) = 0$$

$$AB = 1$$

Soln:

If we add another pole on left side S.T

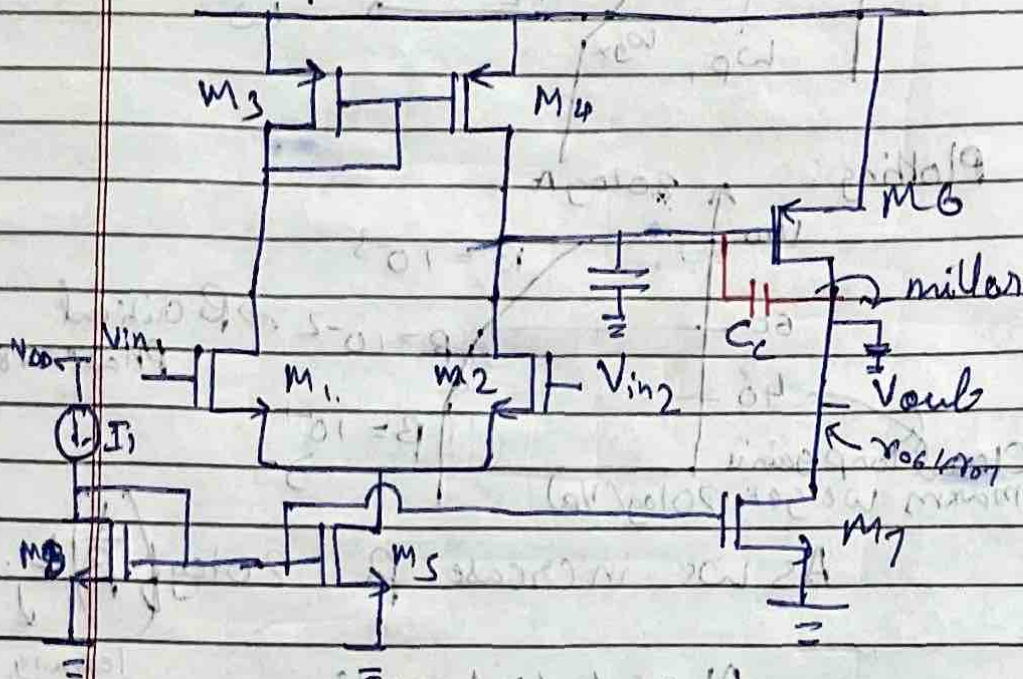


Disadvantage: Bandwidth decreases drastically by doing this

8/09/24

(8A) 0105

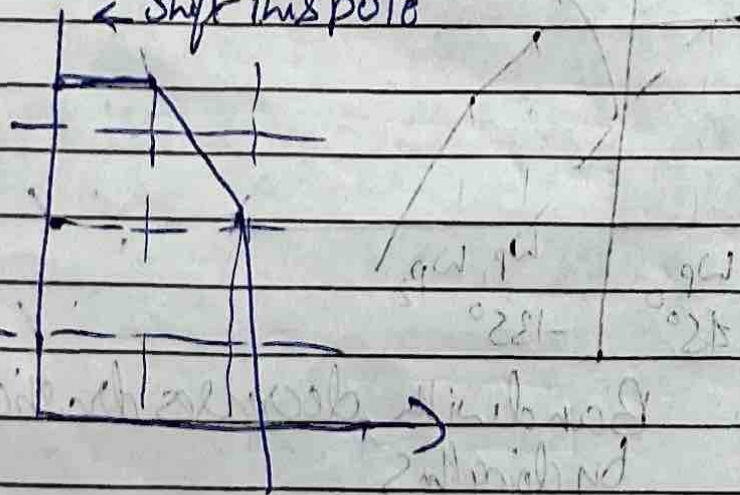
Consider: Two Stage amplifier



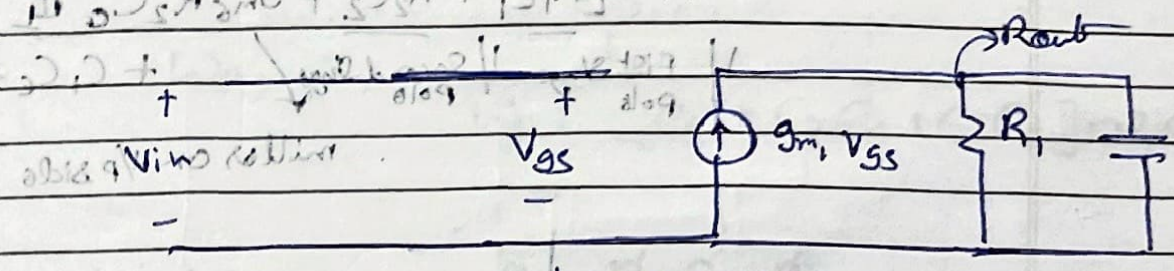
Q. (8A) 0105
1 = 8A

Techniques to increase stability:

← Shift this pole



→ For Pmos I_{CMR} is towards gnd
→ For nmos I_{CMR} is towards V_{dd}

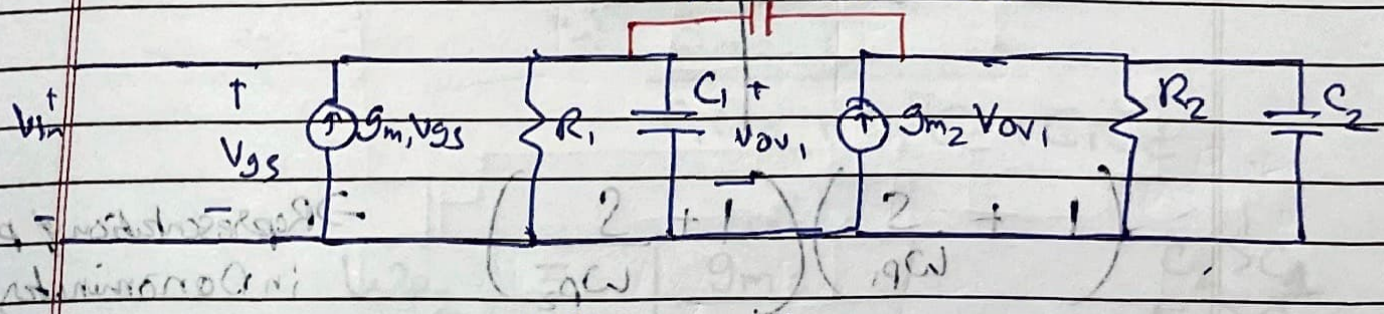


1st stage

Combined Stage:

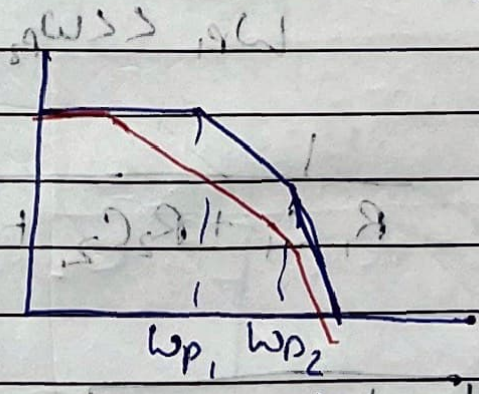
$$R_1 = r_{o2} \parallel r_{o4}$$

$$R_2 = r_{o6} \parallel r_{o7}$$



$$\omega_{p1} = \frac{1}{R_1 C_1}$$

$$\omega_{p2} = \frac{1}{R_2 C_2}$$



$$135^\circ < \phi < 180^\circ$$

For shifting pole 1 $\uparrow C_1$

Due to C_c :

$$V_{out}(s) = \frac{g_{m2} g_{m6} R_1 R_2 C_1 (1 + s C_c / g_{m6})}{V_{in} [1 + s(R_1 C_1 + R_2 C_2 + g_{m6} R_2 C_c R_1) + s^2(C_1 C_c + C_2 C_c) R_1 R_2]}$$

1/ First Stage pole 1/ Second Stage pole

Miller on 1/p side

$$\left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)$$

→ Representation of poles in Denominator

$$\Rightarrow 1 + s \left(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}} \right) + s^2 \frac{1}{\omega_{p1} \omega_{p2}}$$

$$\omega_{p1} \ll \omega_{p2}$$

$s \rightarrow$

$$\therefore \omega_{p1} = \frac{1}{R_1 C_1 + R_2 C_2 + g_{m6} R_2 C_c R_1}$$

$$\approx \frac{1}{g_{m6} R_2 C_c R_1}$$

$$\omega_{p2} = \frac{1}{\omega_{p1}}$$

For $s^2 \rightarrow \dots R_1 R_2 [C_1 C_2 + C_2 C_c + C_1 C_c]$

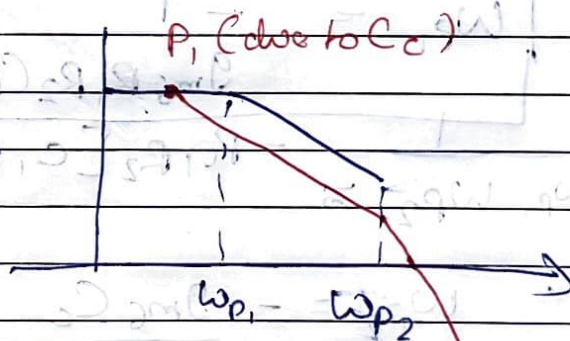
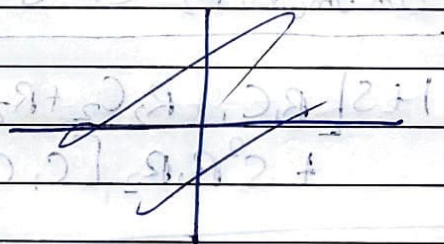
$$\omega_{p2} = \frac{1}{\omega_{p1} [C_1 C_2 + C_2 C_c + C_1 C_c] R_1 R_2}$$

$$\omega_{p2} = \frac{g_{m6} R_2 C_c R_1}{[C_c C_1 + C_2 C_1 + C_c C_2] R_1 R_2}$$

$$\Rightarrow \frac{g_{m6} C_c}{C_1 C_2 + C_2 C_c + C_1 C_c}$$

$$\omega_{p2} = \boxed{\frac{g_{m6}}{C_c}} \quad \begin{matrix} C_2 > C_1 \\ C_c > C_1 \end{matrix}$$

Now !

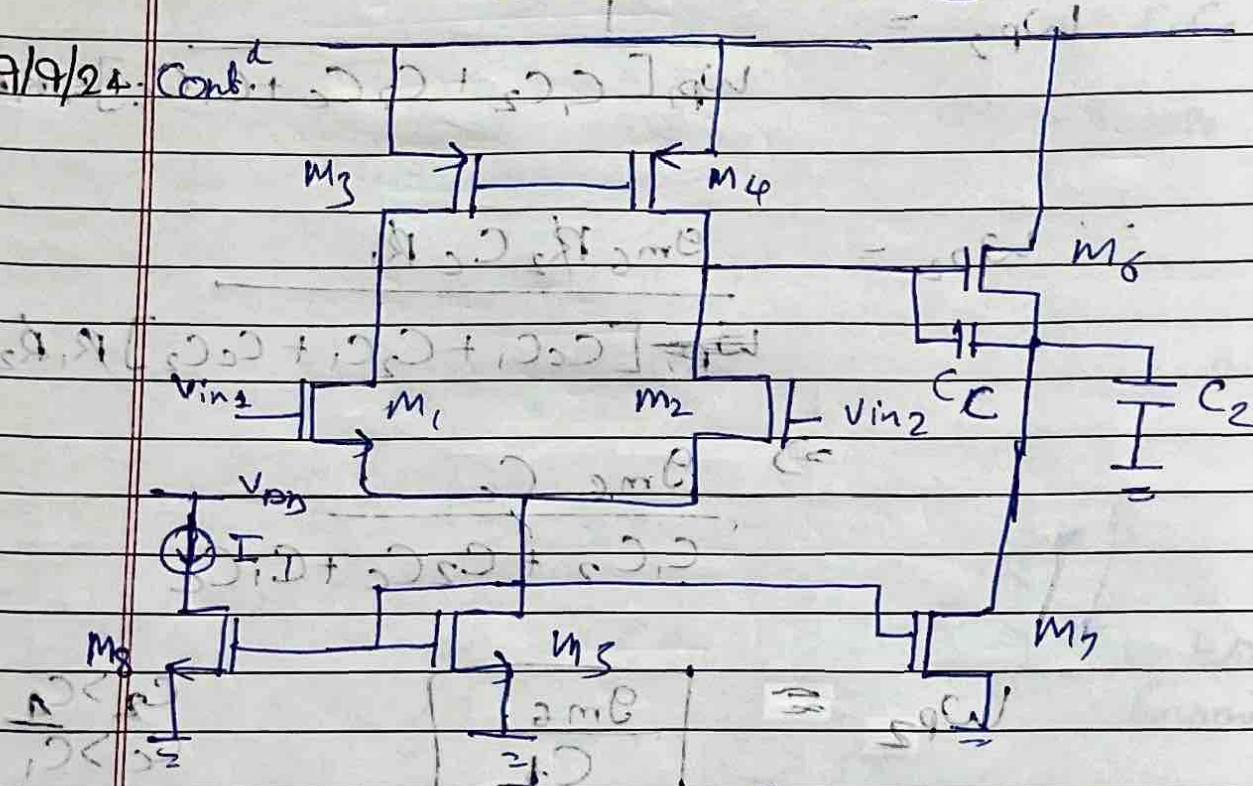


→ improved phase margin

→ ω_{z1} shifted more towards origin

Note: Design the amplifier such that ω_{zero} is far away on Right of ω_{po} to maintain Stability

9/9/24 Cont.



$$\frac{V_{out}(s)}{V_{in}} = \frac{g_m g_{m6} R_1 R_2 (1 - s C_c / g_{m6})}{1 + s [R_1 C_1 + R_2 C_2 + R_2 C_c + (1 + g_{m6} R_2) C_c R_1] + s^2 R_1 R_2 [C_1 C_2 + C_1 C_c + C_2 C_c]}$$

$$\omega_{p1} = \frac{-1}{g_{m6} R_1 R_2 C_c}$$

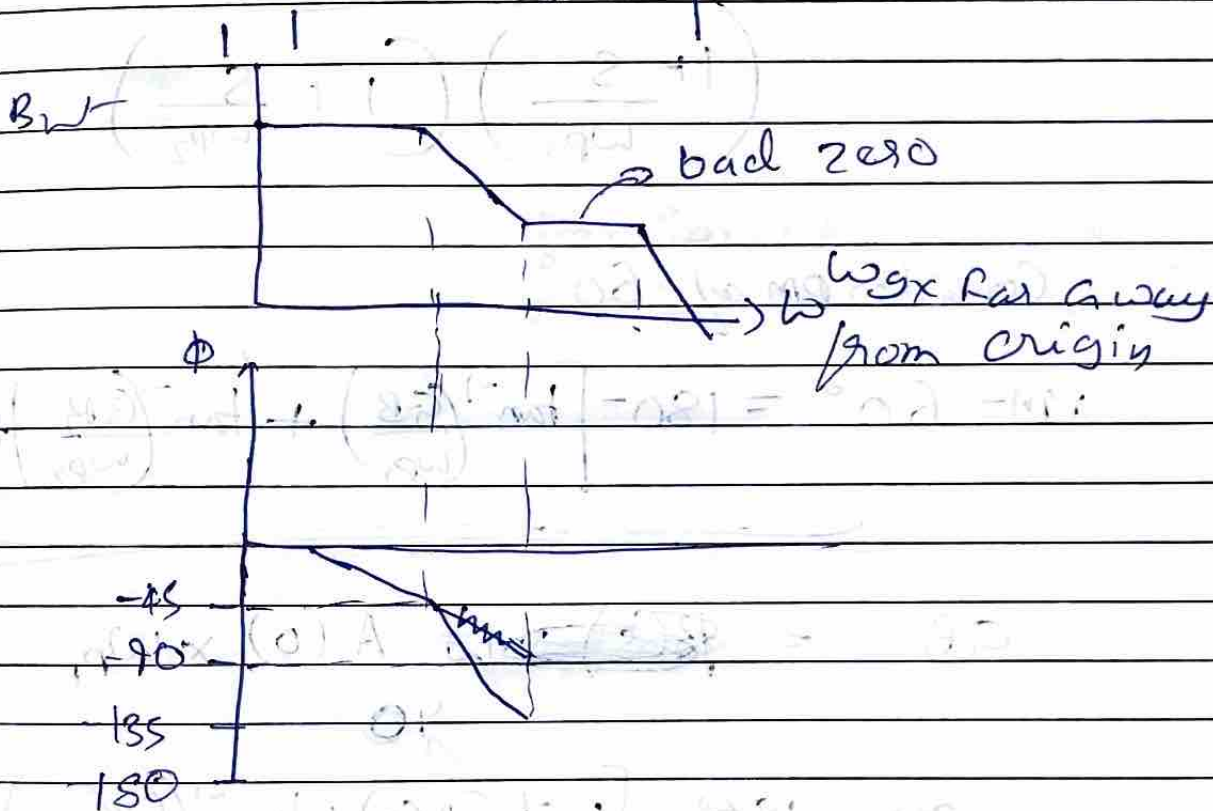
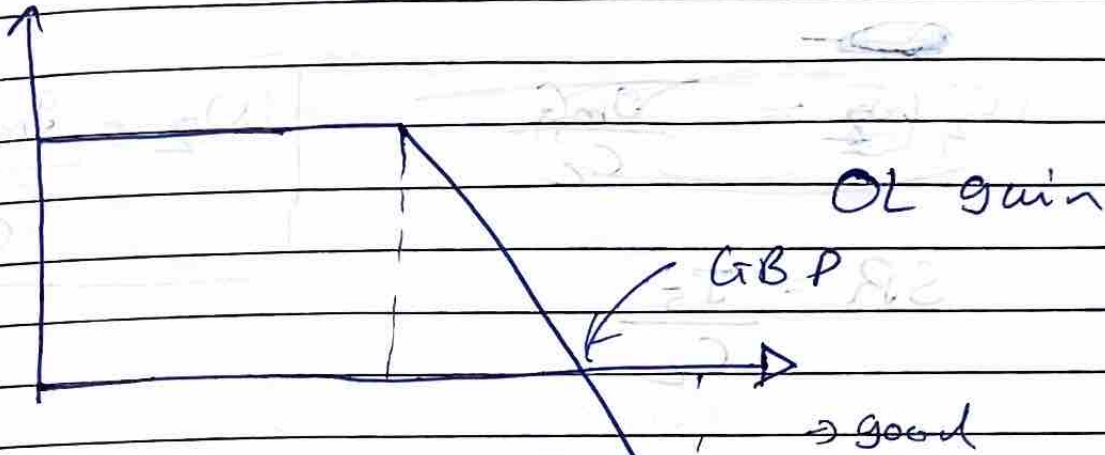
$$\omega_{p1}, \omega_{p2} = R_1 R_2 [C_1 C_2 + C_1 C_c + C_2 C_c]$$

$$\omega_{p2} = \frac{-g_{m6} C_c}{C_1 C_2 + C_1 C_c + C_2 C_c}$$

$$\approx \frac{g_{m6}}{C_2}$$

If $C_c \uparrow$ $\omega_{p2} \uparrow$ $\omega_{p1} \downarrow$

As ω_{p1} & ω_{p2} are close when C_c is absent &
 if C_c is introduced & $C_c \uparrow$ poles separate
 $\rightarrow$ Pole splitting phenomenon



make '0' at 10 times ω_{p1}

$$\boxed{w_2 > 10GBP}$$

$$\omega_2 = \frac{g m c}{c_c}$$

$$SR = \frac{I_5}{C_L}$$

$$\frac{1}{(1 - s/\omega_{p1}) (1 - s/\omega_{p2})}$$
$$PM = 60^\circ = 180 - \left[\tan^{-1} \left(\frac{GB}{\omega_{p1}} \right) + \tan^{-1} \left(\frac{GB}{\omega_{p2}} \right) + \tan^{-1} \left(\frac{GB}{\omega_z} \right) \right]$$

$$GB = \cancel{B(s)} \rightarrow A(s) \times W_p$$

$$\therefore \text{pm} = 180^\circ - \left[\tan^{-1} \left(\frac{A(\omega)}{1} \right) + \tan^{-1} \left(\frac{GB}{\omega \omega_{p2}} \right) + \tan^{-1} (0.1) \right]$$

$$PM = 180 - \left[\tan^{-1} \left(\frac{GB}{\omega_{p2}} \right) \right]$$

$\rightarrow 30^\circ$

$$\therefore \frac{GB}{\omega_{p2}} = \tan (24.28^\circ)$$

$$\therefore \Rightarrow \frac{GB}{\omega_{p2}} = 0.45$$

$$\omega_{p2} = 2.21 GB$$

$$\frac{g_{m6}}{C_L} = 2.2 \times \frac{g_{m2}}{C_C} \rightarrow GB$$

$$\frac{g_{m6} R_{D2} C_C}{\omega_{p2}} \times \frac{g_{m2} g_{m8} R_{D2}}{g_{m2}} = \frac{g_{m2}}{C_C}$$

$$2) \frac{g_{m6}}{C_C} > 10 \frac{g_{m2}}{C_C} \quad \frac{g_{m2}}{C_C} > 2.2$$

$$\therefore C_C > \frac{2.2}{10} C_L$$

$$C_C > 0.22 C_L$$

$$\therefore \frac{g_{m6}}{C_C} > 10 \frac{g_{m2}}{C_C}$$

$$g_{m6} > 10 g_{m2}$$

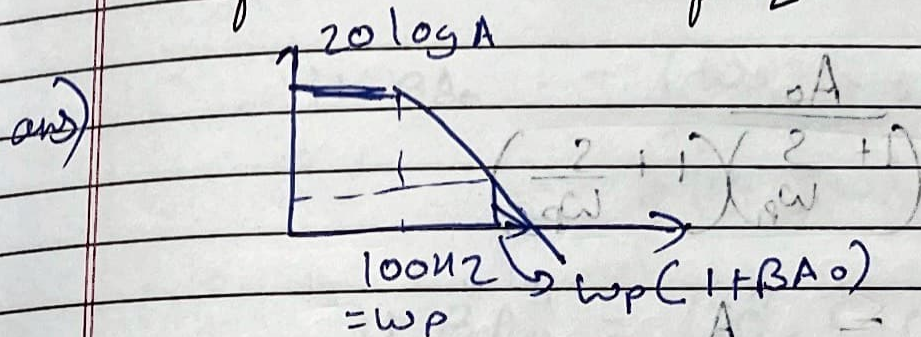
$$\text{As } Q_{mc} = 10 \text{ gm}^2$$

$$2) \quad \frac{10 \text{ gm}^2}{C_L} > \frac{2.2 \text{ gm}^2}{C_C} = 119$$

$$\boxed{C_C > \frac{2.2 C_L}{10}}$$

Tute - 2

- Q1) An op Amp having a single pole rolloff at 100kHz and a low frequency gain of 10^5 is operated in a feedback loop with $\beta = 0.01$. What is the factor by which feedback shifts the pole? To what frequency? If β is changed to a value that results in a nominal closed loop gain of +1 to what freq does pole shift?



New pole location = $100 (1 + B A_0)$

$\rightarrow 100 (1 + 0.01 \times 10^5)$

$\Rightarrow 100 (1001) \text{ Hz}$
 $\Rightarrow \underline{\underline{100.1 \text{ kHz}}}$

If nominal cl gain = $+1 = \frac{1}{\beta} = 1$, $\beta = 1$

New pole location

$= 100 (1 + B A_0)$
 $= 100 (1 + 10^5)$
 $= 10 \text{ MHz}$

Q) Amp with DC gain ≈ 100 , poles at 10^4 & 10^6 rad/s incorporated with negative feedback loop with factor β . For what β does poles coin circle? What is the corresponding Q of the resulting second order system? For what value of β is a maximally flat response achieved? What is the low frequency c/gain in maximally flat case?

Ans $A(s) = \frac{A_0}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)}$

$$A_f(s) = \frac{A}{1 + \beta A(s)}$$

$$A_f(s) = \frac{A_0}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right) + \beta A_0} = \frac{\omega_{p1} \omega_{p2} A_0}{(s + \omega_{p1})(s + \omega_{p2}) + \beta A_0 \omega_{p1} \omega_{p2}}$$

or $A_f(s) = \frac{A_0}{1 + s\left(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}}\right) + \frac{s^2}{\omega_{p1}\omega_{p2}} + \beta A_0}$

$$A_f(s) = \frac{A_0 \omega_{p1} \omega_{p2}}{s^2 + s(\omega_{p1} + \omega_{p2}) + (\beta A_0 + 1) \omega_{p1} \omega_{p2}}$$

$$s^2 + s(\omega_{p1} + \omega_{p2}) + (\beta A_0 + 1) \omega_{p1} \omega_{p2}$$

Note: DC gain $= A_0$

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Denominator $= 0 \Rightarrow$

$$S = \frac{-(\omega_{p1} + \omega_{p2}) \pm \sqrt{(\omega_{p1} + \omega_{p2})^2 - 4(1 + \beta A_0)\omega_{p1}\omega_{p2}}}{2}$$

For poles to coincide $\Rightarrow$

$$(\omega_{p1} + \omega_{p2})^2 - 4(1 + \beta A_0)\omega_{p1}\omega_{p2} = 0$$

$$1 + \beta A_0 = \frac{(\omega_{p1} + \omega_{p2})^2}{4\omega_{p1}\omega_{p2}}$$

$$1 + \beta A_0 = \frac{(10^4 + 10^6)^2}{4 \times 10^{4+6}} = 25.5025$$

$$\beta = \frac{25.5025 - 1}{100} = 0.245025$$

Comparing denominator with: $s^2 + \omega_0 s + \omega_0^2 = 0$

$$\text{we get } \omega_0 = \sqrt{(1 + \beta A_0)\omega_{p1}\omega_{p2}}$$

$$Q = \frac{\omega_0}{\omega_{p1} + \omega_{p2}}$$

$$\therefore Q = \frac{\sqrt{(1 + \beta A_0)\omega_{p1}\omega_{p2}}}{\omega_{p1} + \omega_{p2}} = 0.5$$

For maximally Flat response

$$|A(f)| = \frac{1}{\sqrt{2}} = \frac{\sqrt{(1 + BA_0) \omega_{p1} \omega_{p2}}}{\omega_{p1} + \omega_{p2}}$$

$$(\omega_{p1} + \omega_{p2})^2 = 2(1 + BA_0) \omega_{p1} \omega_{p2}$$

$$\therefore 1 + BA_0 = \frac{(\omega_{p1} + \omega_{p2})^2}{2 \omega_{p1} \omega_{p2}} = 51.005$$

$$B = \frac{51.005 - 1}{100} = \underline{\underline{0.50005}}$$

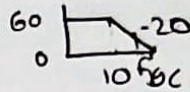
Q3) Consider an op amp having a single pole, OL response with $A_0 = 10^5$ & $f_p = 10 \text{ Hz}$. Let the op amp be ideal i.e. ∞ input res, zero output res. If this amplifier is connected in the non-inverting configuration with a nominal low-frequency closed-loop gain of 100, find the frequency at which $|A(f)| = 1$. Also find the phase margin.

ans) $1/B = 100 \Rightarrow B = 0.01$

$$B A(s) = \frac{0.01 \times 10^5}{\left(1 + \frac{s}{20 \times 10}\right)}$$

$$\Rightarrow \frac{60 - 0}{\log\left(\frac{10}{f_{gc}}\right)} = -20$$

$$f_{gc} = 10^4 \text{ Hz}$$



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$$PM = 180 - \tan^{-1} \left(\frac{10^4}{10} \right) = 180^\circ - 89.9427^\circ$$

$$\phi = 90 - 0.573^\circ$$

Q4) Find closed loop gain at ω , relative to low freq gain when $PM = 30^\circ, 60^\circ, 90^\circ$

$$|A_F(\omega)| = \frac{1/B}{\sqrt{(1 + \cos \theta)^2 + \sin^2 \theta}}$$

$$\text{For } PM = 30^\circ \quad \theta = 150^\circ$$

$$|A_F(\omega)| = 19.3 \cdot 18.52$$

$$\text{For } PM = 60^\circ \quad \theta = 120^\circ$$

$$|A_F(\omega)| = 100$$

$$\text{For } PM = 90^\circ \quad \theta = 90^\circ$$

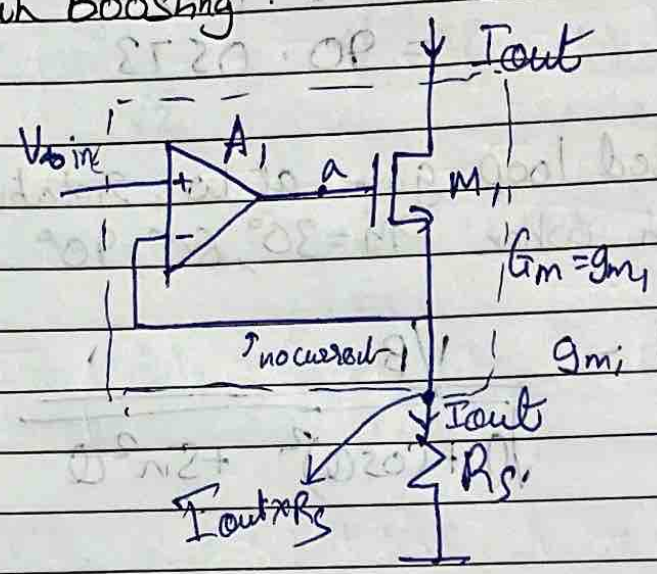
$$A_F(\omega) = 70.71$$

$$A_F(\omega) = 1$$

$$A_F(\omega) = 1$$

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Gain Boosting



$$V_D = g_m V_{GS}$$

$$\rightarrow (V_{in} - I_{out} R_s) A_1 \rightarrow p.t.a$$

$$V_{GS} = [(V_{in} - I_{out} R_s) A_1 - I_{out} R_s] g_{m1}$$

$$\rightarrow I_{out} = (V_{in} - I_{out} R_s) A_1 - I_{out} R_s] g_{m1}$$

$$\rightarrow V_{in} A_1 g_{m1} = I_{out} [1 + A_1 R_s g_{m1} + g_{m1} R_s]$$

$$\frac{I_{out}}{V_{in}} = \frac{g_{m1} A_1}{1 + g_{m1} A_1 R_s + g_{m1} R_s}$$

$$\Rightarrow \frac{g_{m1} A_1}{1 + g_{m1} (1 + A_1) R_s}$$

$$OR G_m = \frac{g_{m1} A_1}{1 + g_{m1} (1 + A_1) R_s}$$

Consider:

Refers V_m

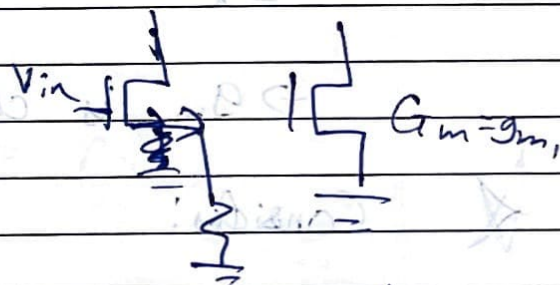
$$G_m = \frac{g_m}{1 + g_m R_S}$$

$$G_m = \frac{g_m}{1 + g_m R_S}$$

$$V = \left(\frac{1}{1 + g_m R_S} \right) g_m R_{out} = g_m (A_1 R_1)$$

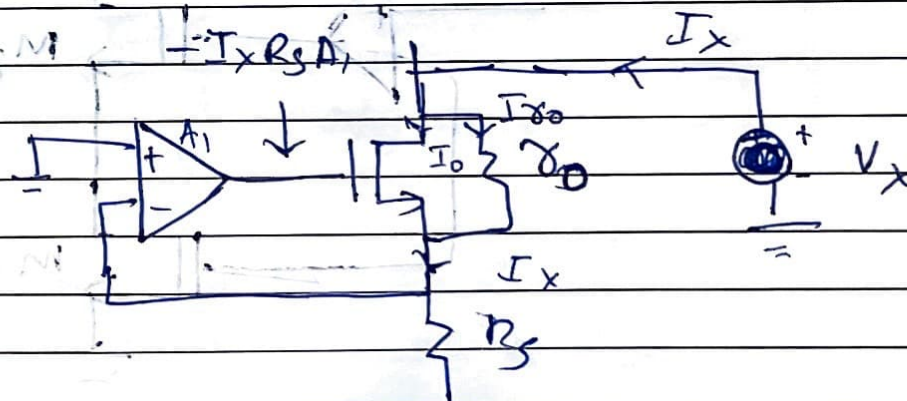
→ This circuit is called superstar.

$$\rightarrow A_v = G_m R_{out}$$



$$G_m = g_m / (1 + g_m R_S)$$

Consider:



$$(-I_x R_S A_1 - I_x R_S) g_m = I_o$$

$$I_{o0} = \frac{V_x - I_x R_S}{r_o}$$

$$I_{x_0} + I_o = I_x$$

$$I_x = (-I_x R_s A_1 - I_x R_s) g_m + \frac{V_x - I_x R_s}{r_o}$$

$$\Rightarrow I_x \left(1 + (A_1 R_s + R_s) g_m + \frac{R_s}{r_o} \right) = \frac{V_x}{r_o}$$

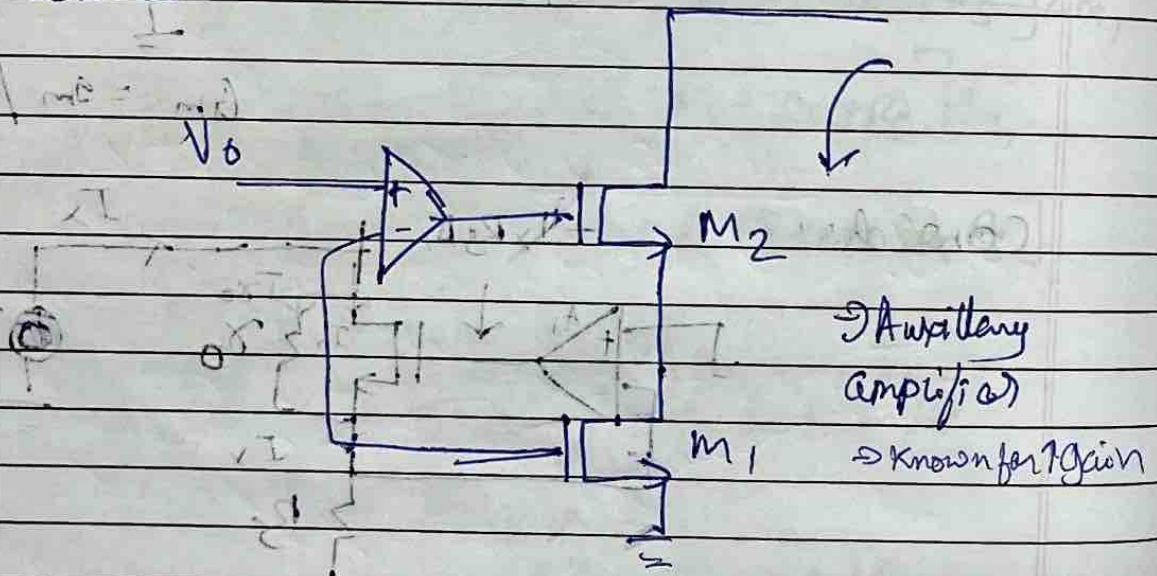
Routoff

$$\frac{V_x}{I_x} = r_o + R_s + (1 + A_1) g_m r_o R_s$$

$\Rightarrow g_m$ is changed again i.e. multiplied by $1 + A_1$



Consider:

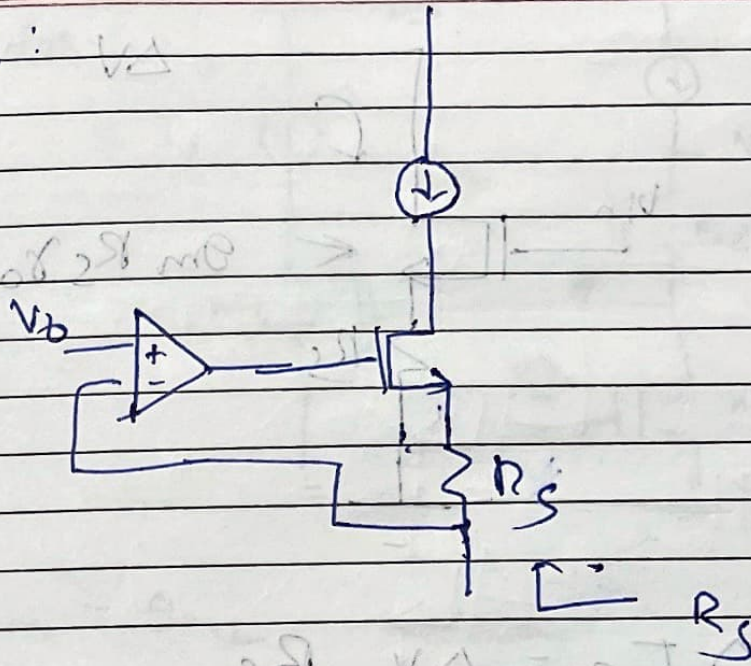


$$R_{out}^{(with A)} = R_s + r_o + g_m R_s r_o$$

$$R_{out}^{(with A)} = r_{o1} + r_{o2} + (1 + A_1) g_{m2} r_s r_{o2}$$

Roout ↑
Gain ↑

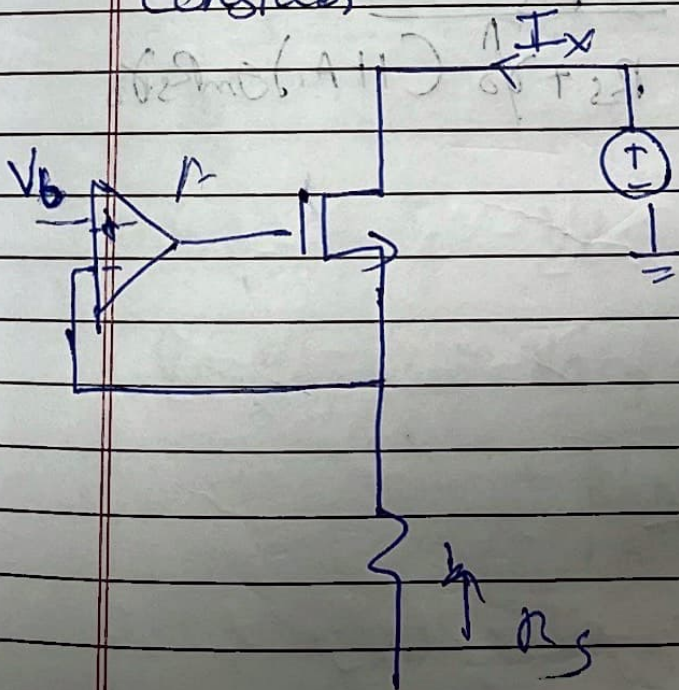
Suppose : V_A



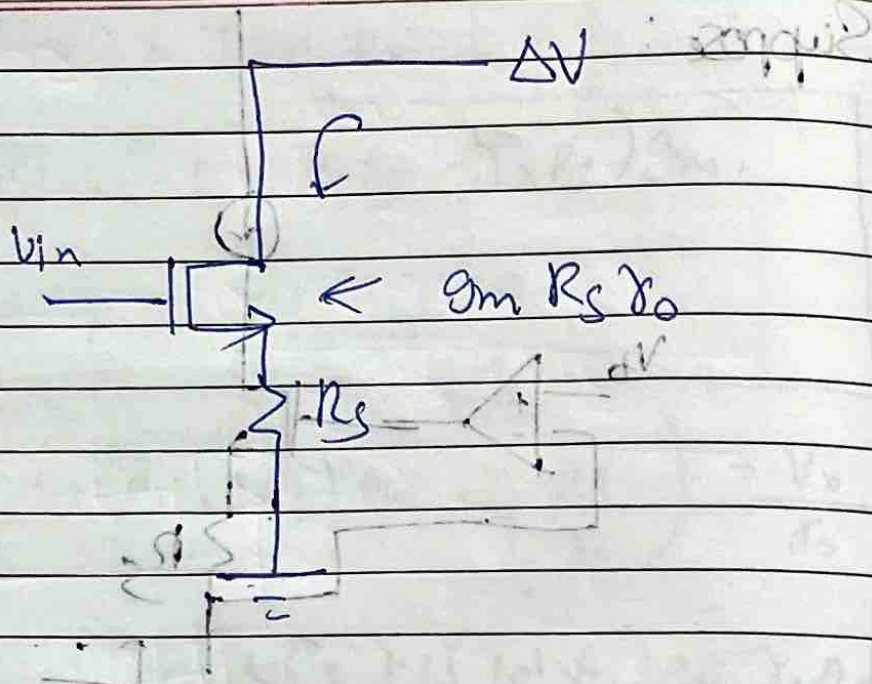
If Ckt is

$$R = R_D + \frac{r_o}{1 + g_m r_o}$$

Consider :



⇒ Same thing



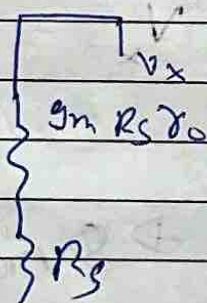
8/13

$$\Delta I_R = \Delta V \frac{R_s}{R_s + r_o + g_m R_s r_o}$$

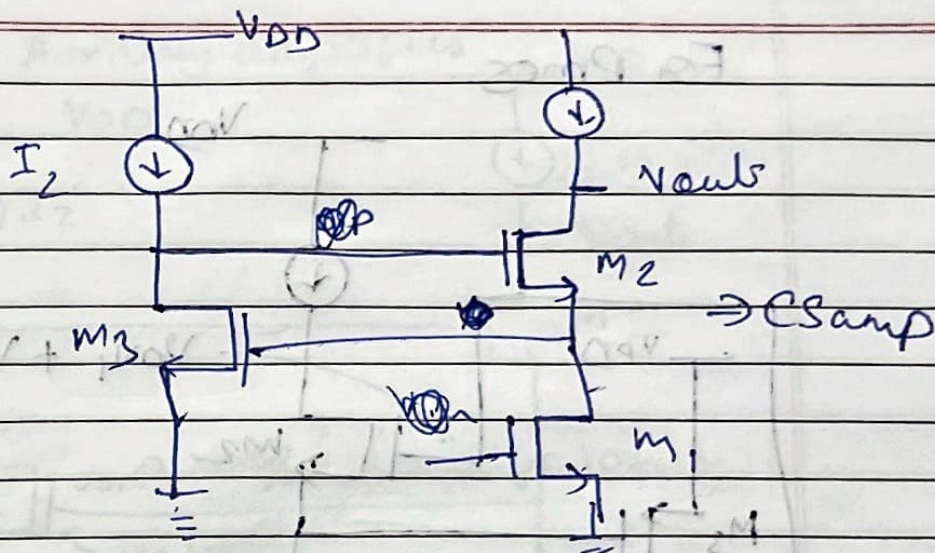
With an amplifier

$$\Delta I_x = \Delta V \times \frac{R_s}{R_s + r_o (1 + A_v) g_m R_s r_o}$$

Too small



Consider :



$$G_m = g_{m2}$$

$$R_{out} = R_{st} \parallel r_{o1} \parallel (1 + A_1) g_{m2} R_g \parallel r_{o2}$$

$$A_v = g_{m2} R_{out}$$

$$A_1 = g_{m3} r_{o3} \quad \rightarrow ? \text{ how}$$

$$\therefore R_{out} = r_{o1} \parallel r_{o2} \parallel (1 + A_1) g_{m2} R_g \parallel r_{o1}$$

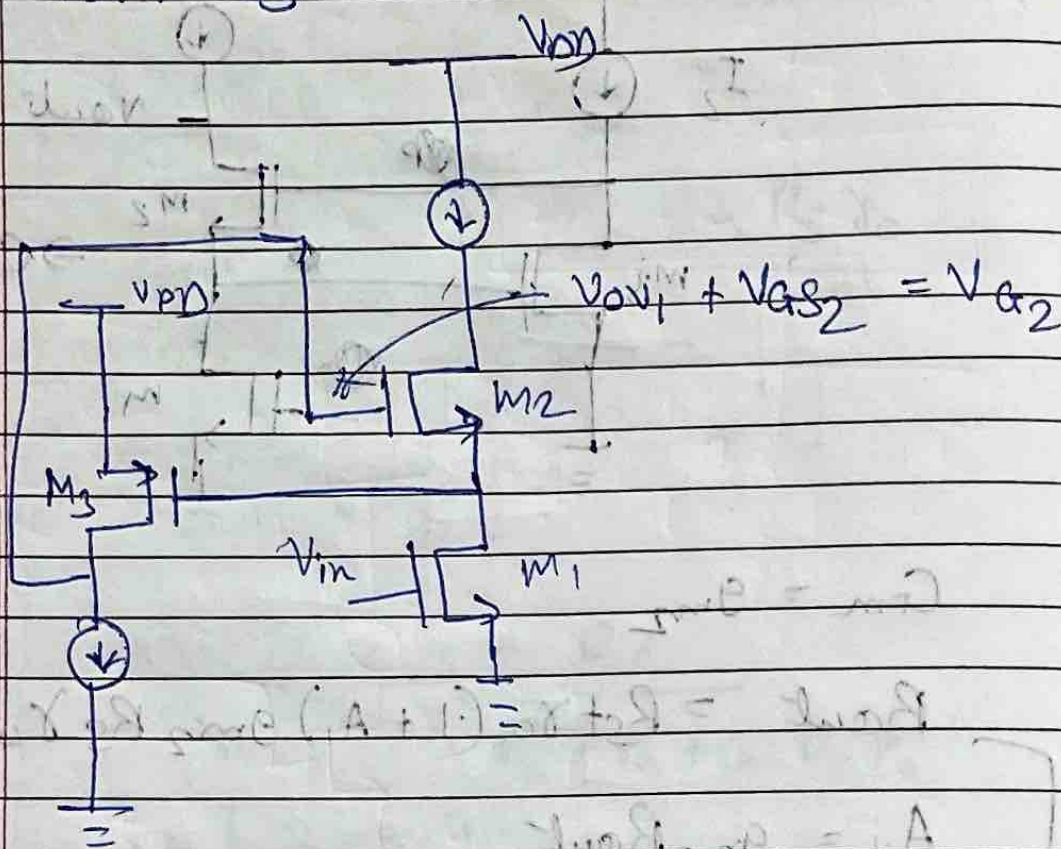
¶

Note: $V_{in} = \frac{1}{2} V_{DD}$ is sine wave. O/P is cosine wave

$$V_{out \min} = V_{GS3} + V_{DS3} + V_{ov2}$$

$$= V_{ov3} + V_{m3} + V_{ov2}$$

For Pmos:



For M3 to be in Sat:

$$V_{GS} > V_{GS} - |V_{th3}|$$

$$V_S - V_D$$

$$\rightarrow V_S - V_{G2} > V_S - V_{D1} - |V_{th3}|$$

$$\rightarrow -(V_{ov1} + V_{GS2}) > -V_{ov1} - |V_{th3}|$$

$$V_{GS2} < |V_{th3}|$$

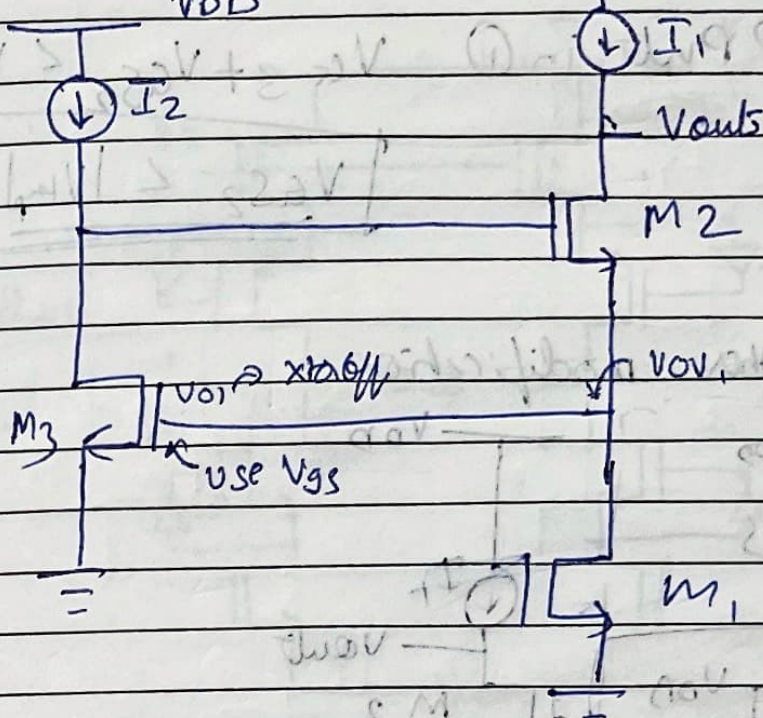
M3 will always be at edge of triode region

V_{ov3} will be very less resulting in high $\left(\frac{W}{L}\right)_3$

Consider:

VDD

VDD



→ Calculate from this side for Volt

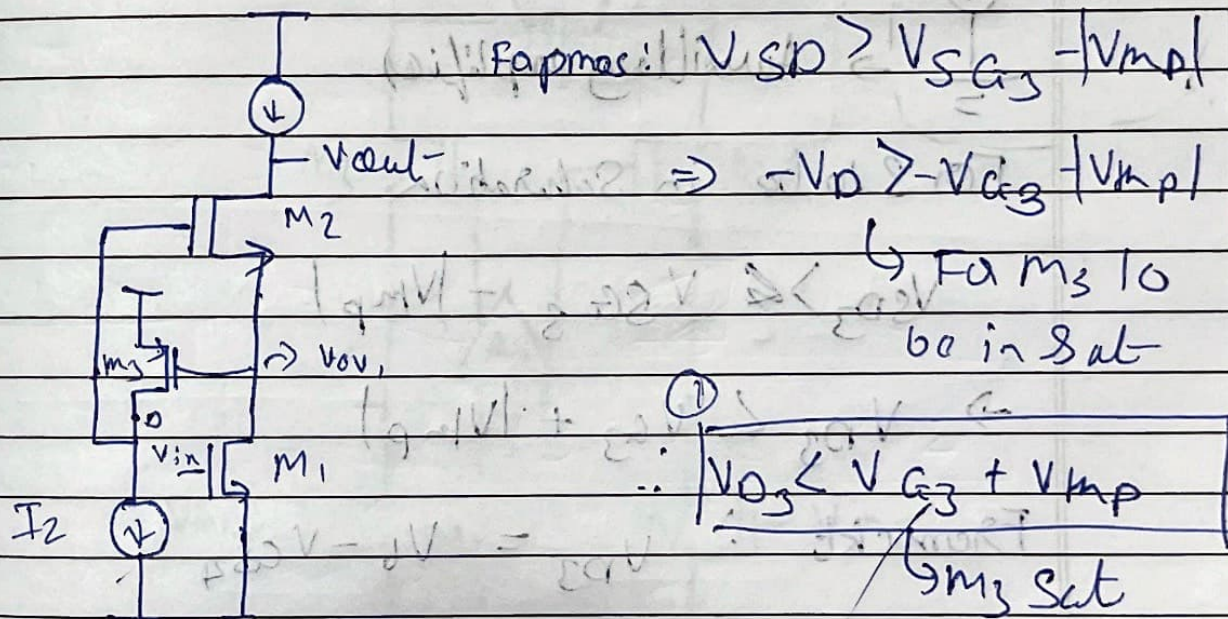
$$V_{out\ min} = V_{GS} + V_{ov2}$$

Modification :

Ex: $V_{SD} > V_{SG} - |V_{mp}|$

$$u_{\text{in}} \Rightarrow -V_D > -V_{D3} + |V_{thp}|$$

↳ Family to be in Sat

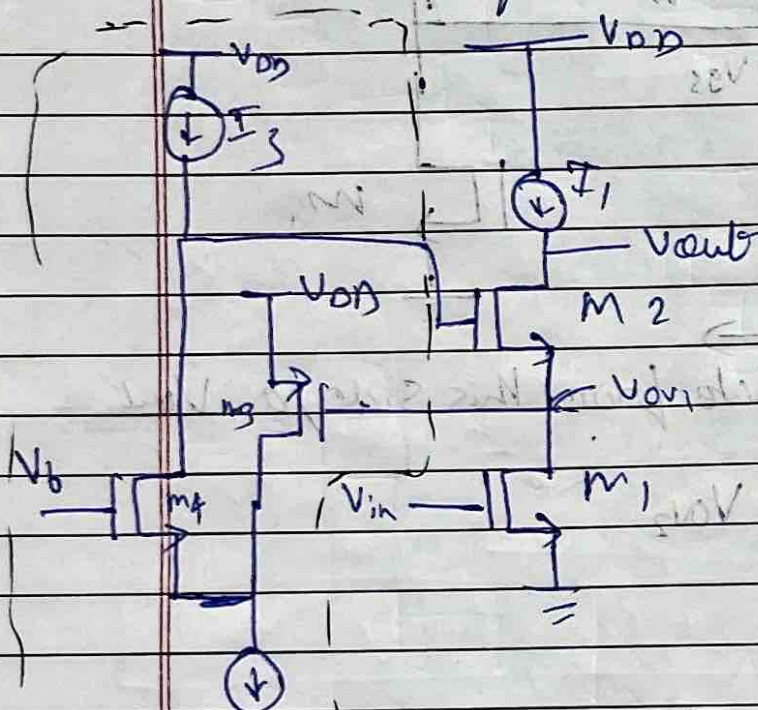


$$V_{D3} = V_{G3} + V_{GS2}$$

Put in ① $V_{G3} + V_{GS2} < V_{G3} + |V_{thp}|$

$$\boxed{V_{GS2} < |V_{thp}|}$$

Further modification



Auxiliary amplifier

For 3 to be in Saturation

$$V_{SD3} > V_{SG3} - |V_{thp}|$$

$$\Rightarrow V_{D3} < V_{G3} + |V_{thp}|$$

Fromckt $V_{D3} = V_b - V_{GS4}$

$$\therefore V_b - V_{GS4} < V_{G3} + |V_{thp}|$$

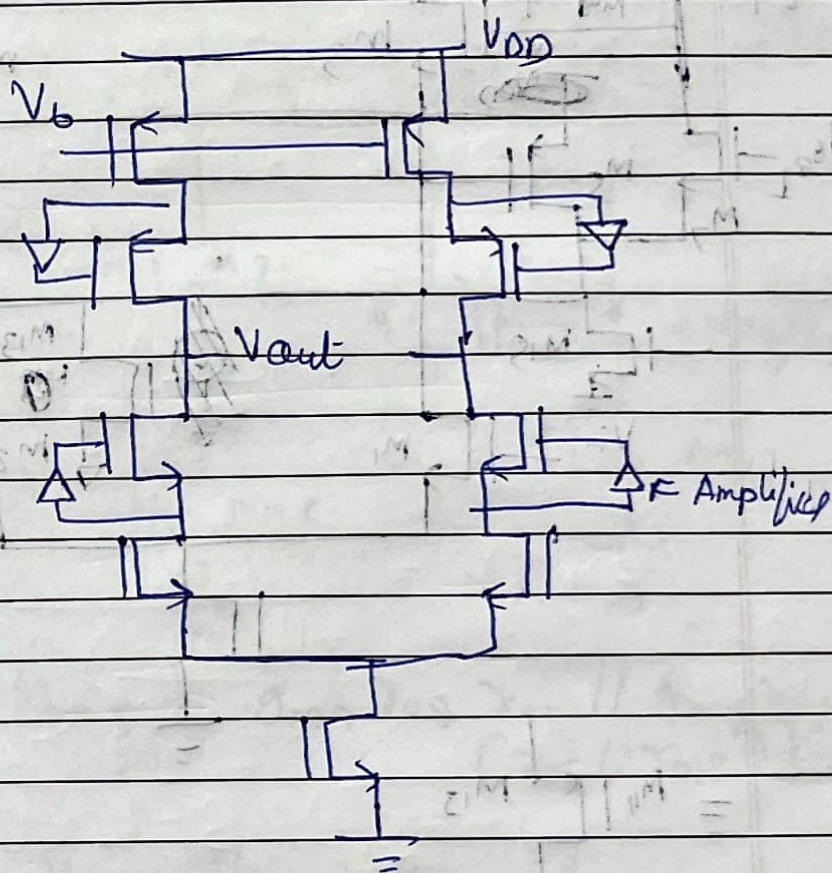
$\hookrightarrow \min \Rightarrow V_{bi}$

Note: If Pmos of input source is connected to ground in AC $G_m = g_m$

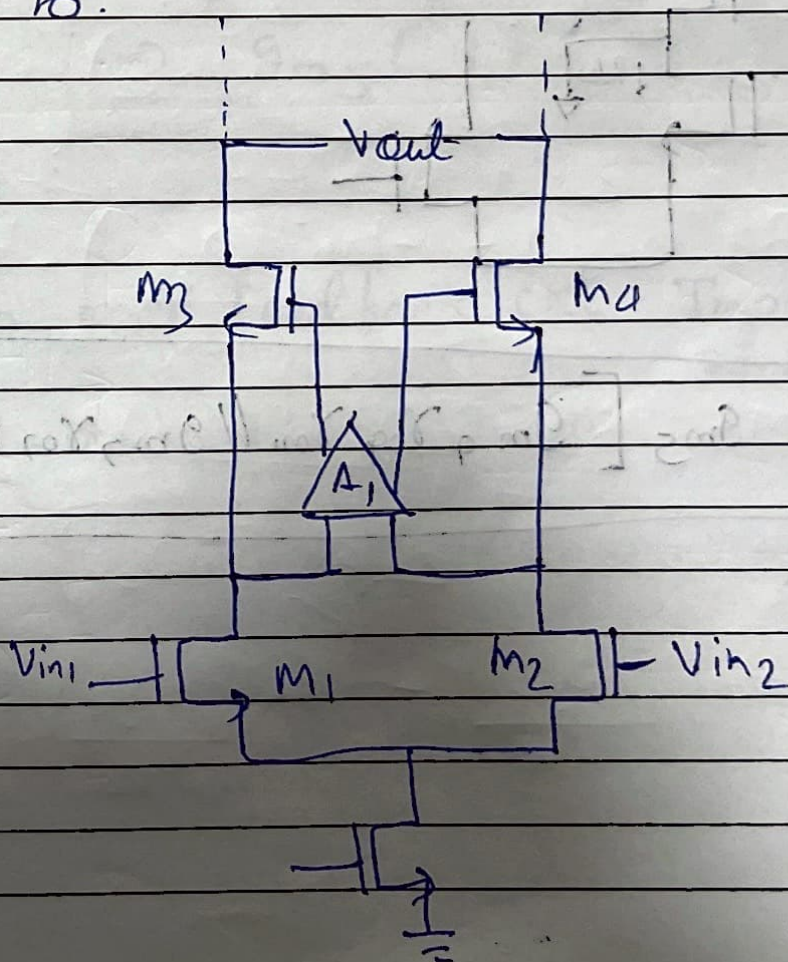
$$A_1 = g_{m3}$$

Consider

To boost
Rout & thus
Gain



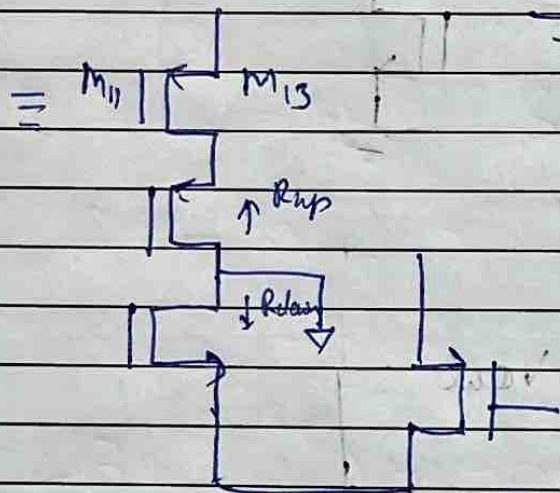
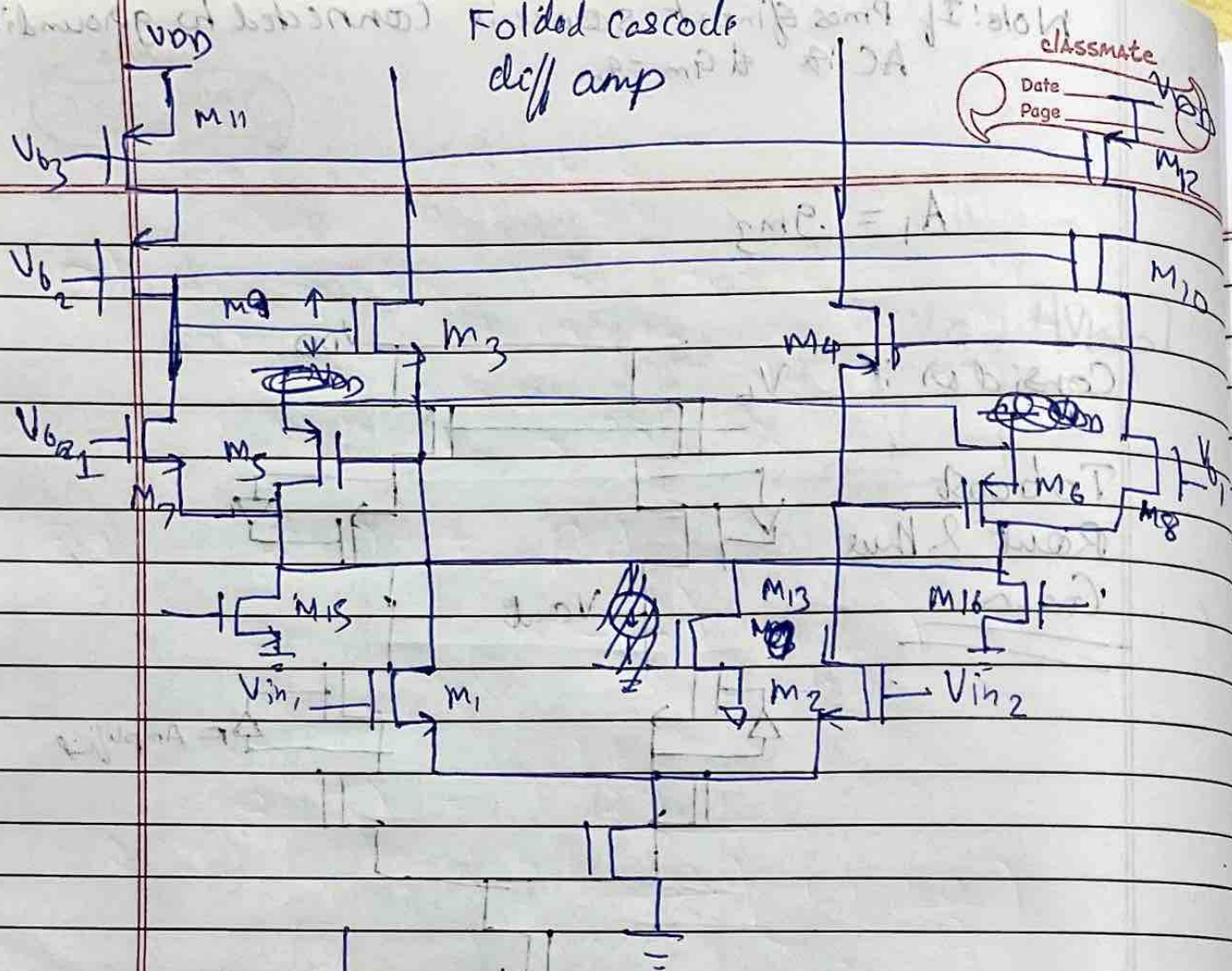
Equivalent to:



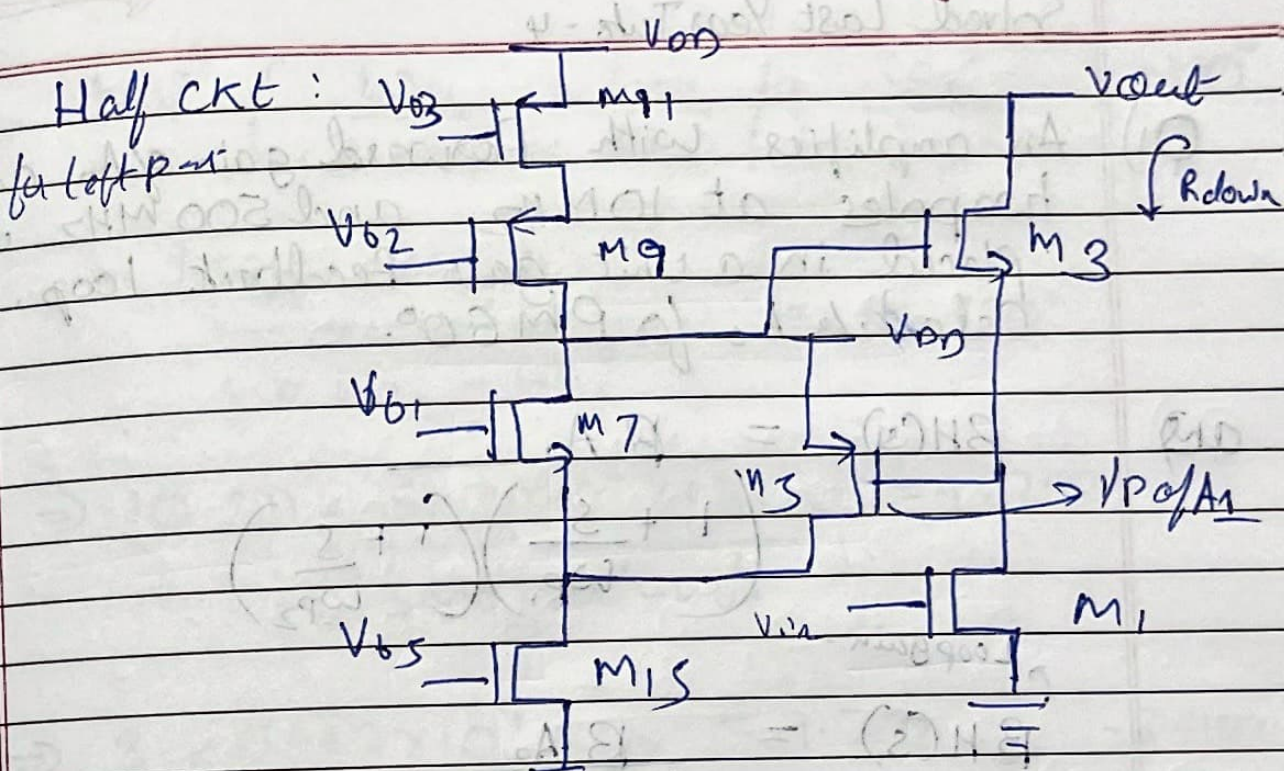
Folded cascode dc/amp

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$$A_1 = g_{m5} [g_{m9} r_{o9} r_{o11} // g_{m7} r_{o7} (r_{o5} // r_{o15})]$$



$$A_1 = g_{m5} \left[g_{m9} r_{o9} r_{o11} \parallel g_{m7} r_{o7} (r_{o5} \parallel r_{o15}) \right]$$

Note: O/P voltage is not changed

$$\rightarrow g_{m5} (1 + A_1) r_{o3} r_{o1} \rightarrow F_{down}$$

~~Consider~~ :

Common mode Feedback CKT: Important for stabilising

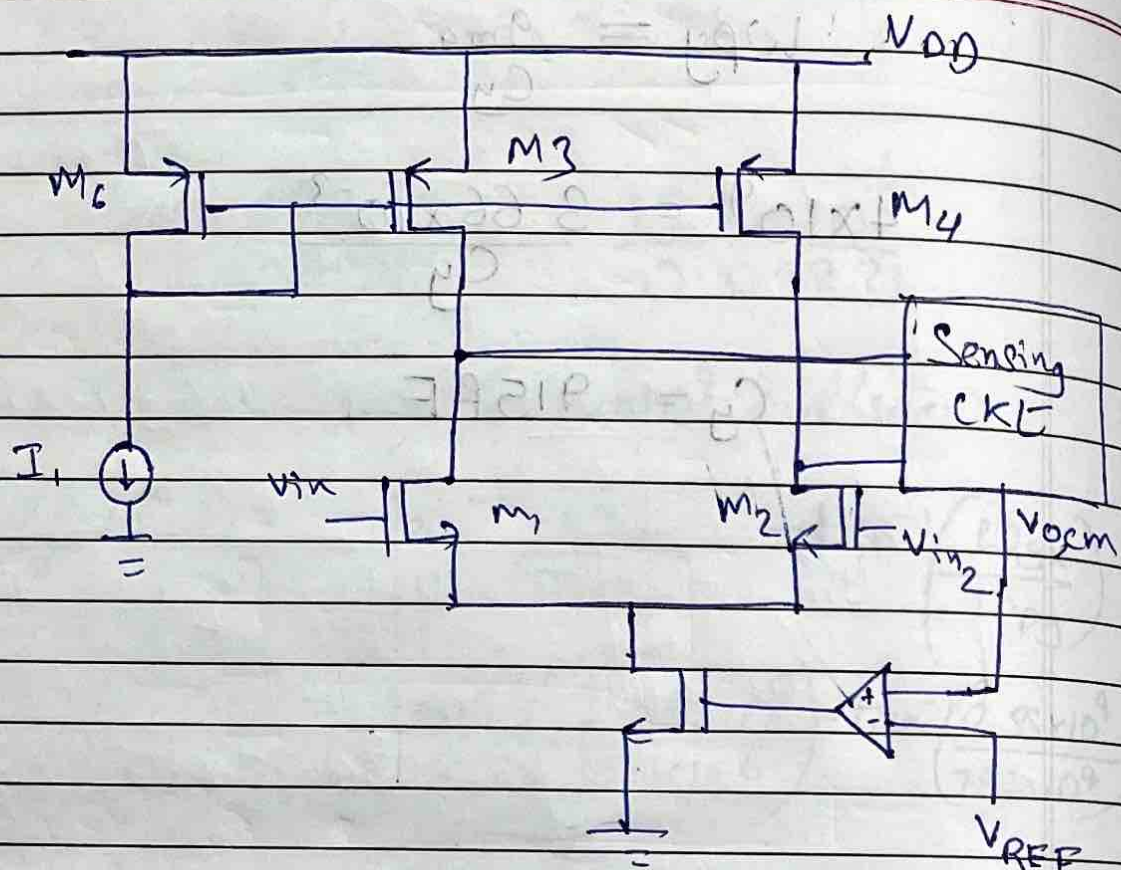
input common mode

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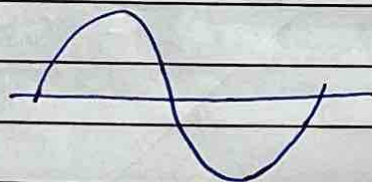
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V_{outcm} should be middle of output common mode range

$$V_{outcm,avg} \approx \frac{V_{outmax} + V_{outmin}}{2}$$

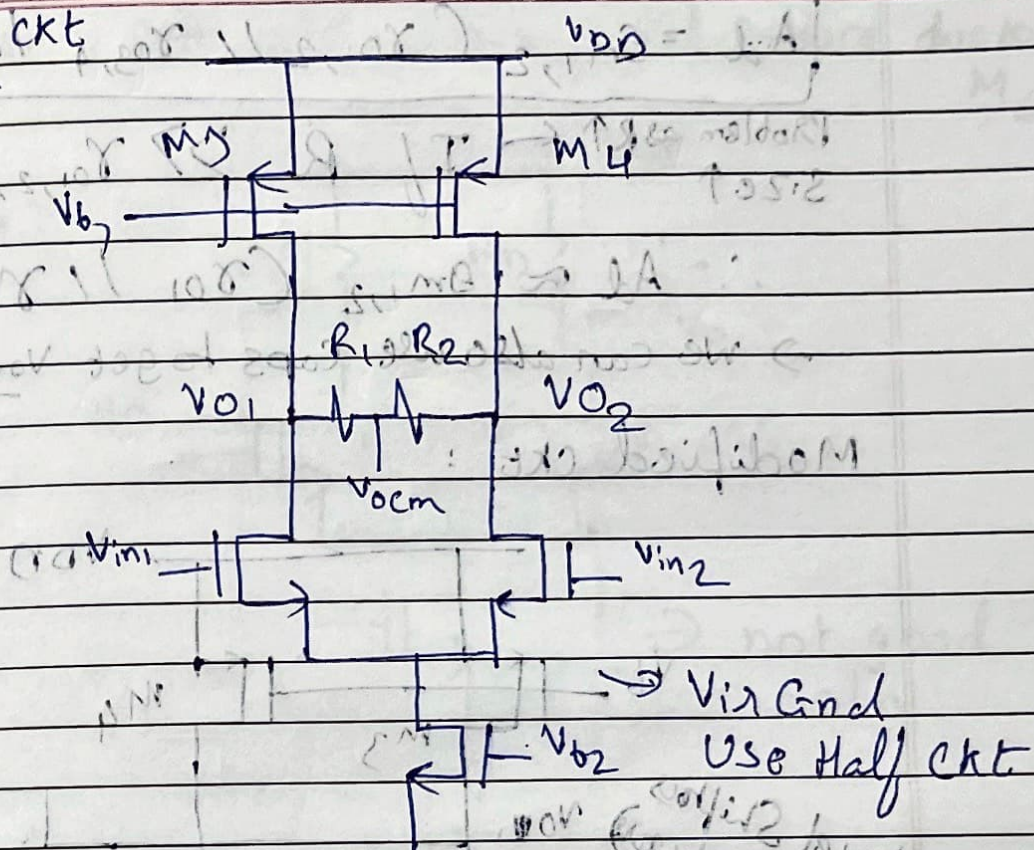
Egi



0.2

Sensing ckt $\rightarrow$ Senses common mode voltage

$$\text{Common mode I/P voltage} = \frac{V_{in1} + V_{in2}}{2}$$

Simple ckt
demo

$$V_{outcm} = \frac{V_{out1} R_2}{R_1 + R_2} + \frac{V_{out2} R_1}{R_1 + R_2}$$

Principle
of superposition

$$V_{outcm} = \frac{V_{out1} R_2 + V_{out2} R_1}{R_1 + R_2}$$

$$\text{if } R_1 = R_2 = R$$

$$V_{outcm} = \frac{V_{out1} + V_{out2}}{2}$$

$$0.2V = \frac{0.1V + 0.1V}{2} = 0.1V$$

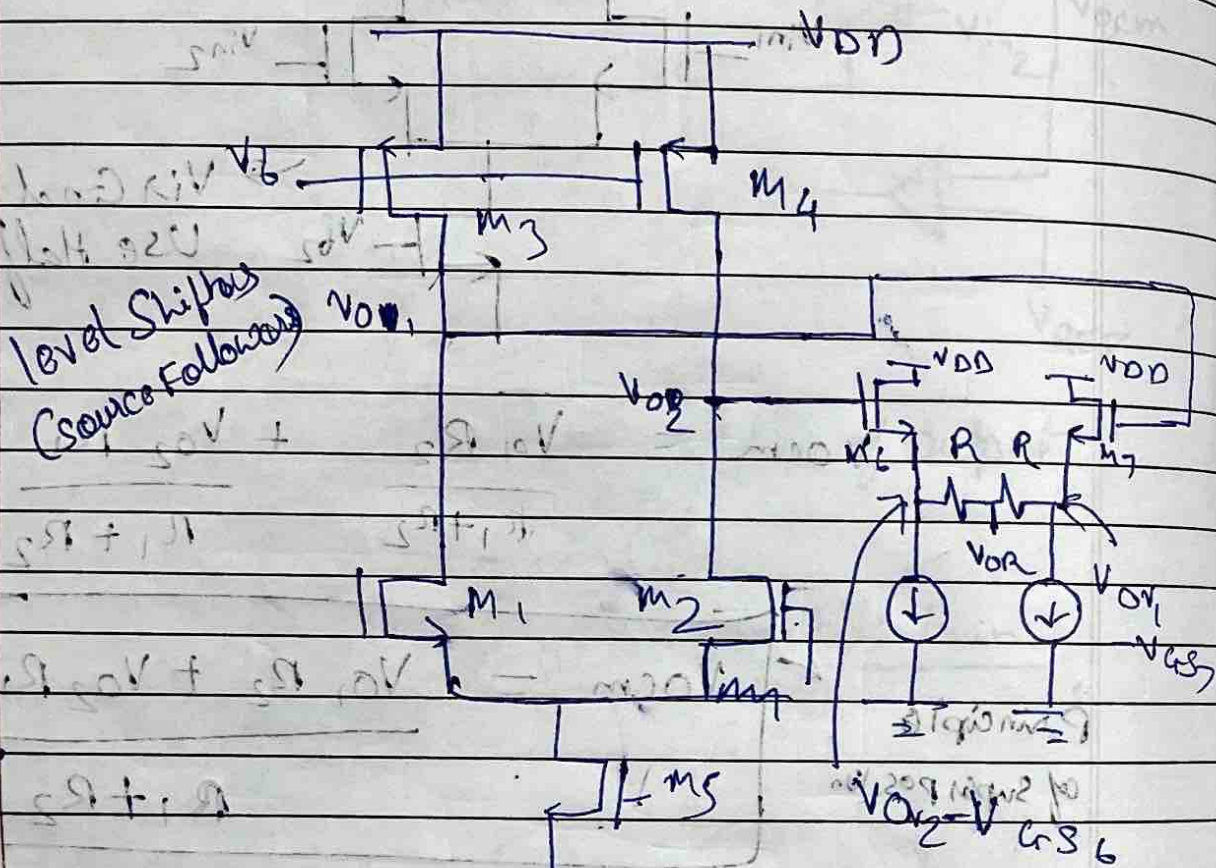
$$A_d = g_{m1,2} (r_{o1,2} \parallel r_{o3,4} \parallel R_{eq})$$

Problem as $R_T \leftarrow I/R \gg r_{o1,2,3}$
size $\uparrow$

$$\therefore A_d \approx g_{m1,2} (r_{o1} \parallel r_{o3})$$

$\rightarrow$ we can also use caps to get $\frac{V_{ov1} + V_{ov2}}{2}$ but BWN

Modified ckt :

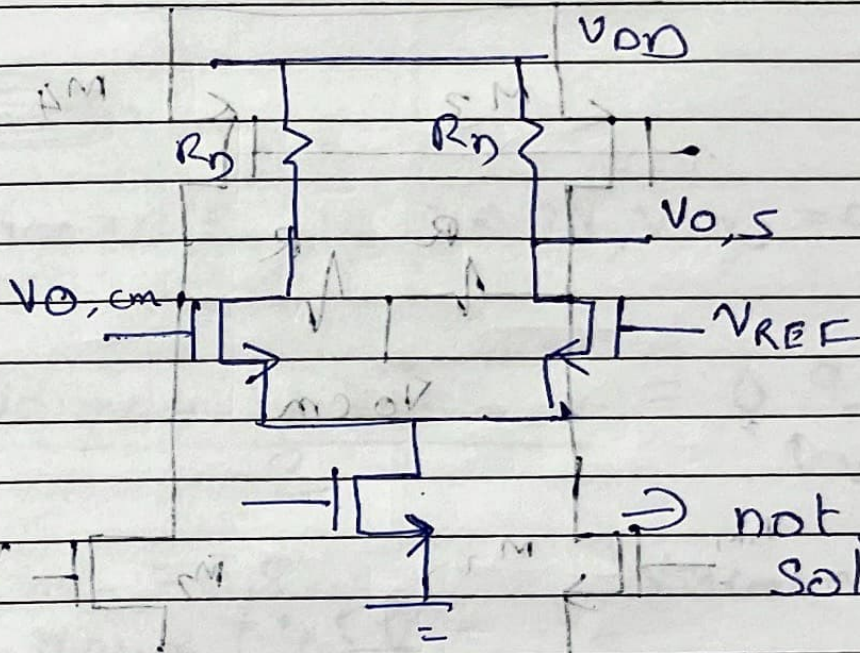


$$V_{ov} = \frac{V_{ov1} + V_{ov2}}{2} = \frac{V_{gs7} + V_{ov2} - V_{gs6}}{2}$$

$$V_{ov} = \frac{V_{ov1} + V_{ov2} - V_{gs6}}{2}$$

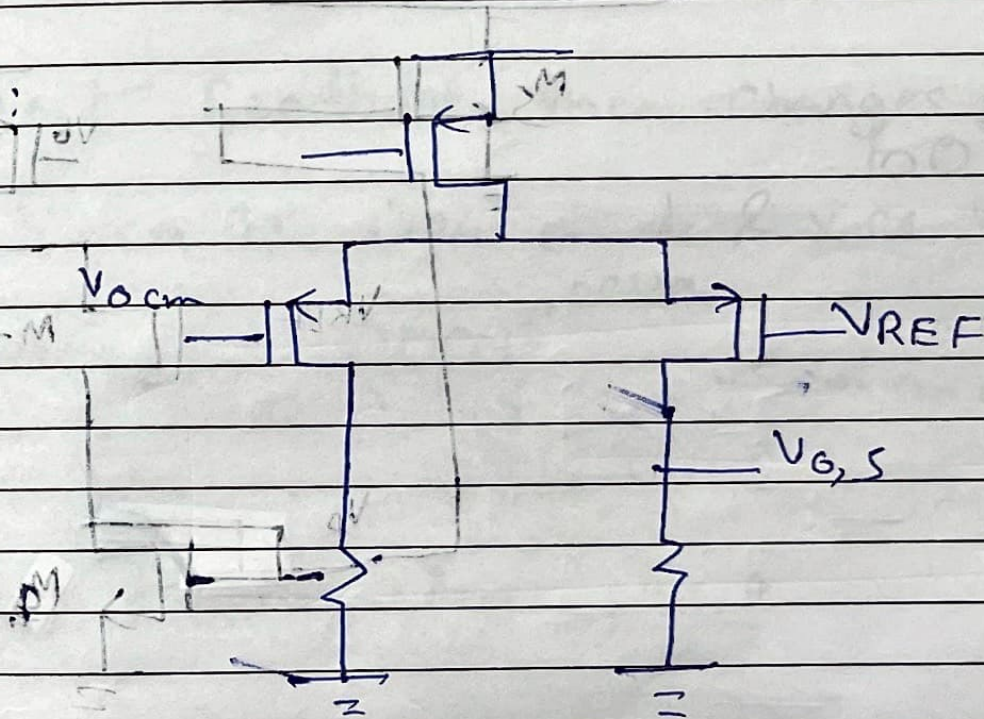
Problem: O/P voltage swing will reduce due to M_6

Consider!

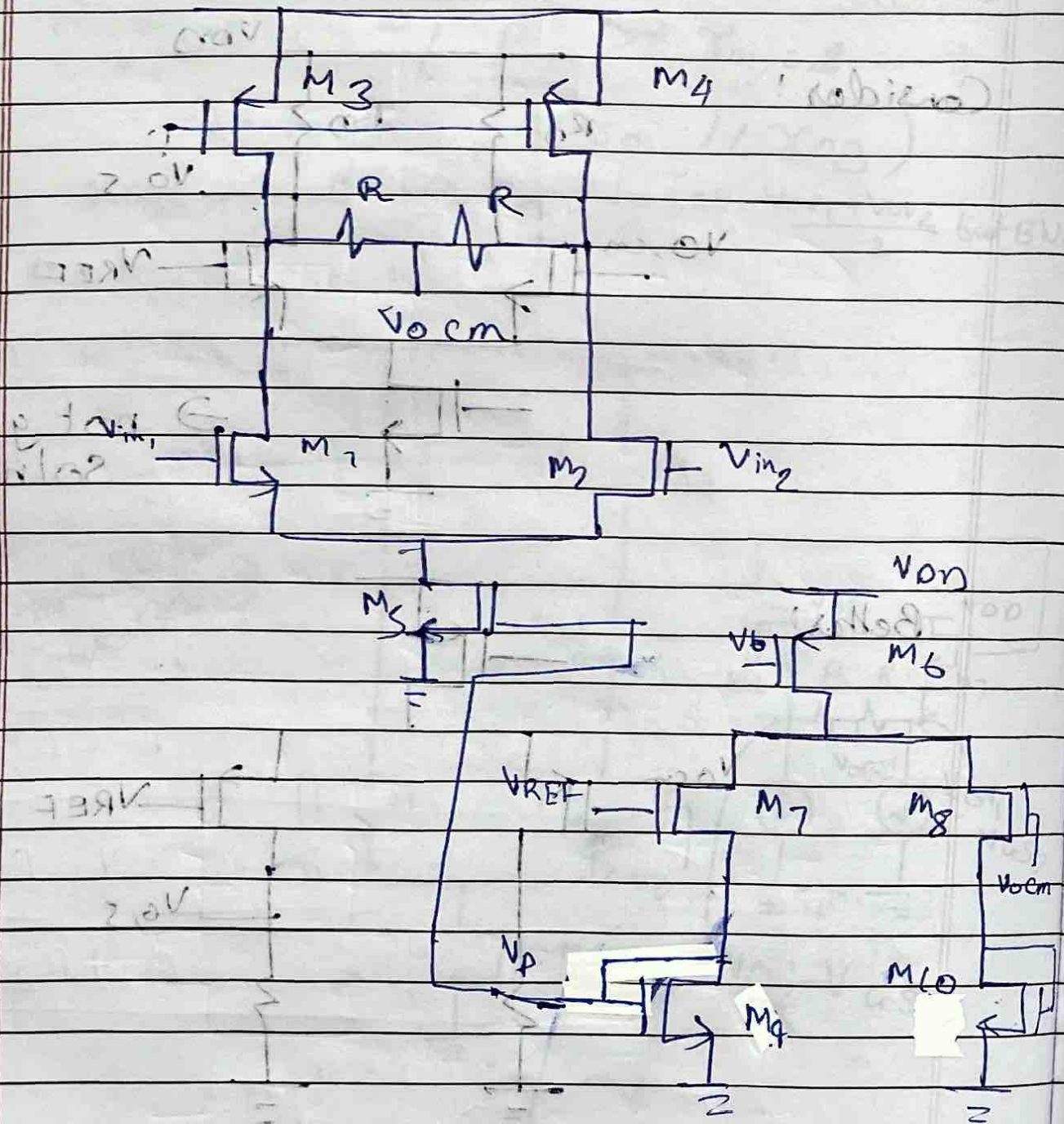


⇒ not good soln

Better:



Final Solution (Best)



Due to V_{OCM} , V_P

V_{REF} , V_P

$V_{GS} - V_{th} = 0.8$
 $\rightarrow 0.1$
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 $GS = 0.6$

$\rightarrow V_{O,cm} = 0.95 \pm 0.1\%$ when $V_{in,cm}$ changes from min to max

$V_{in,cm}$

Eg: Let $V_{DD} = 1.8$, $V_b = 0.2V$, $V_{OV,S} = 0.2V$
 $V_{OV,1} = 0.1V$

$\Rightarrow \frac{V_{O,max} + V_{O,min}}{2} = 0.95$
 $\uparrow V_{O,cm,mid}$

$V_{in,cm,min} = 0.8V$, $V_{in,cm,mid} = 1.25V$
 $V_{in,cm,max} = 1.45V$

$\Rightarrow$ Without feedback $V_{O,cm}$ changes from 0.3 to 0.6V

$\rightarrow$ If $V_{in,cm} \uparrow$, $V_{out,cm}$ drops vice versa

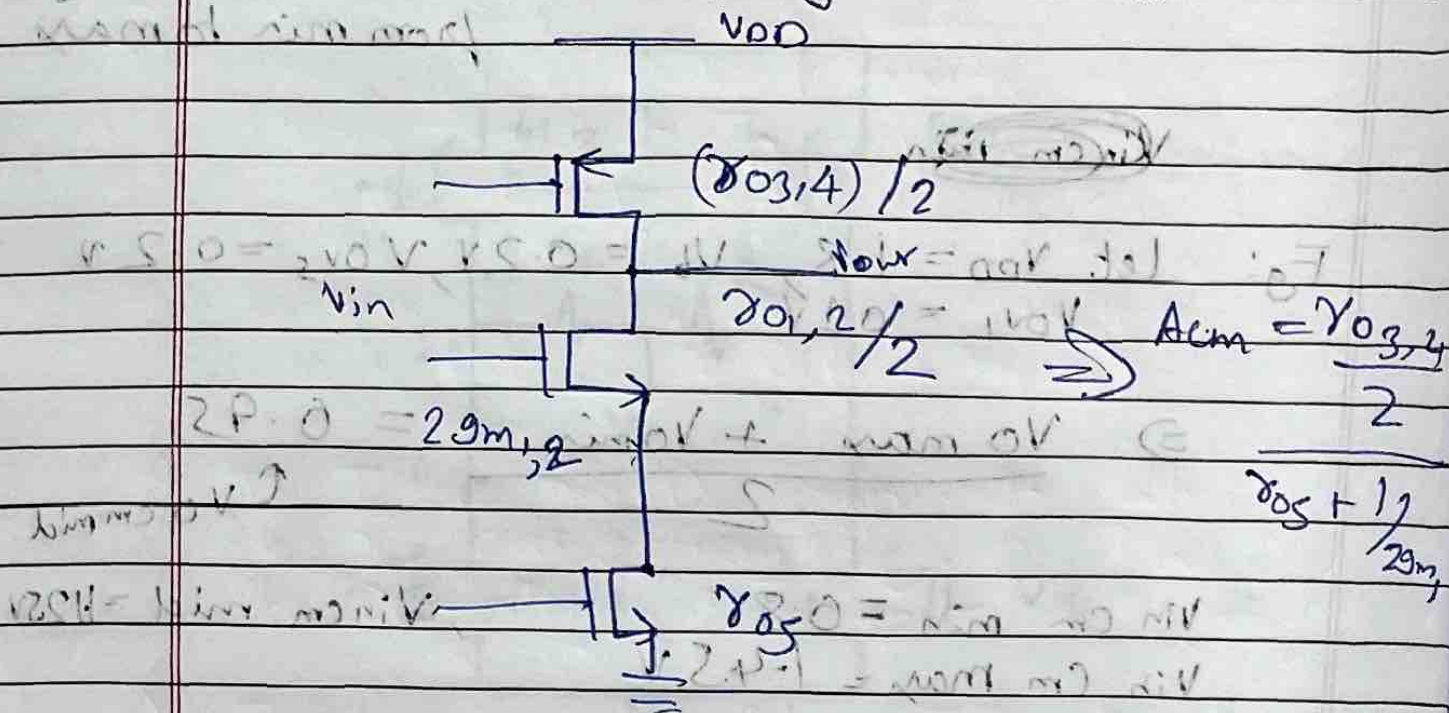
$\Rightarrow A_{cm} = \frac{0.00045}{0.325} = 0.002923$
 $\rightarrow 1.25 - 0.8$
 $\rightarrow V_{in,cm,mid}$

$\Rightarrow$ Without feedback $\rightarrow A_{cm} = \frac{\gamma_{os,4}}{2}$

$\gamma_{os} + \frac{1}{2g_m}$

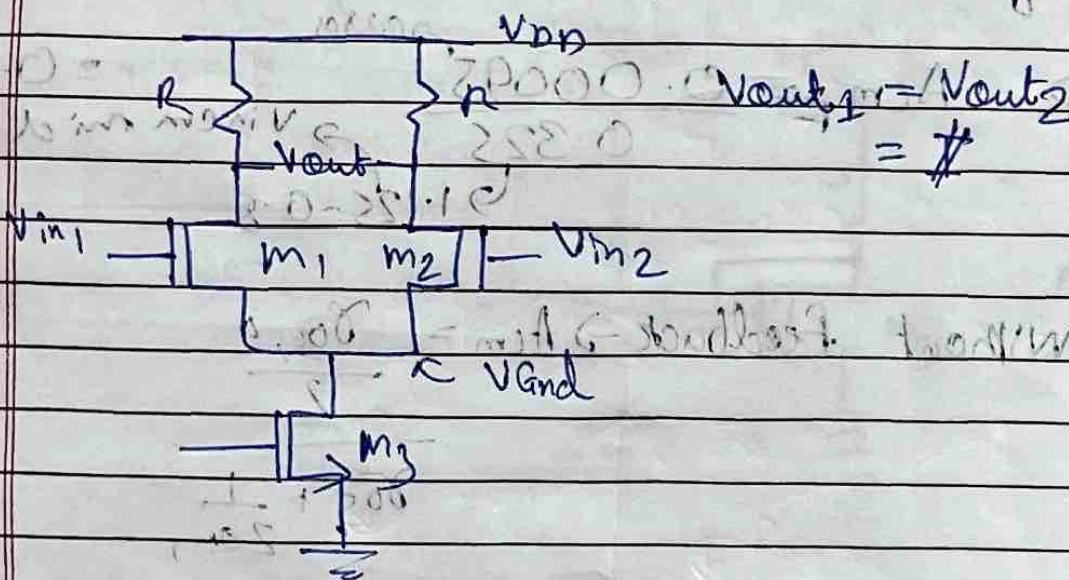
$10.6 \times 922T - 0.01 = 10.6 \times 922T - 0.01$

For common mode gain calculation



Missed morning class

Common Mode Feedback (CMFB)

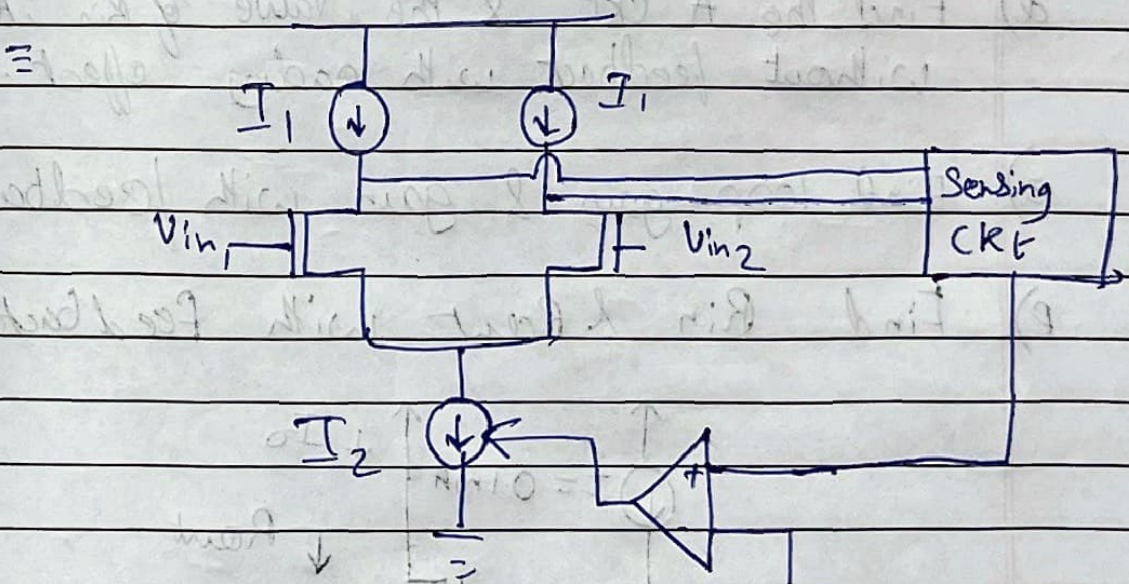
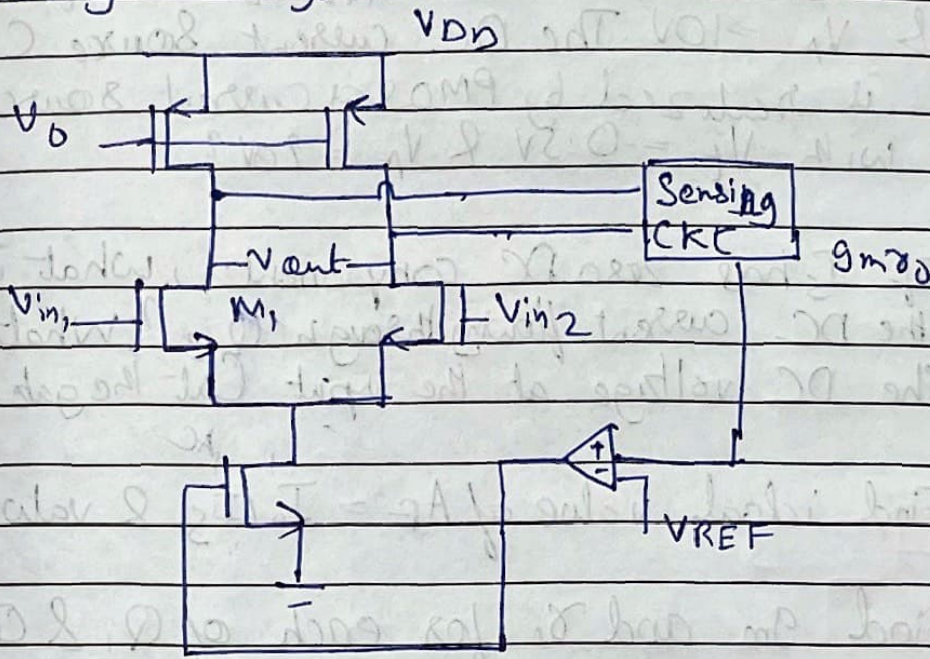


$$V_{out1} = V_{out2} = V_{DD} - \frac{I_{SS} R_D}{2} \rightarrow \text{O/P CM level}$$

$$A_v = -g_m (R_o \parallel r_o) \approx -g_m R_o$$

(if $R_o \ll r_o$)

Sensing ckt wage!



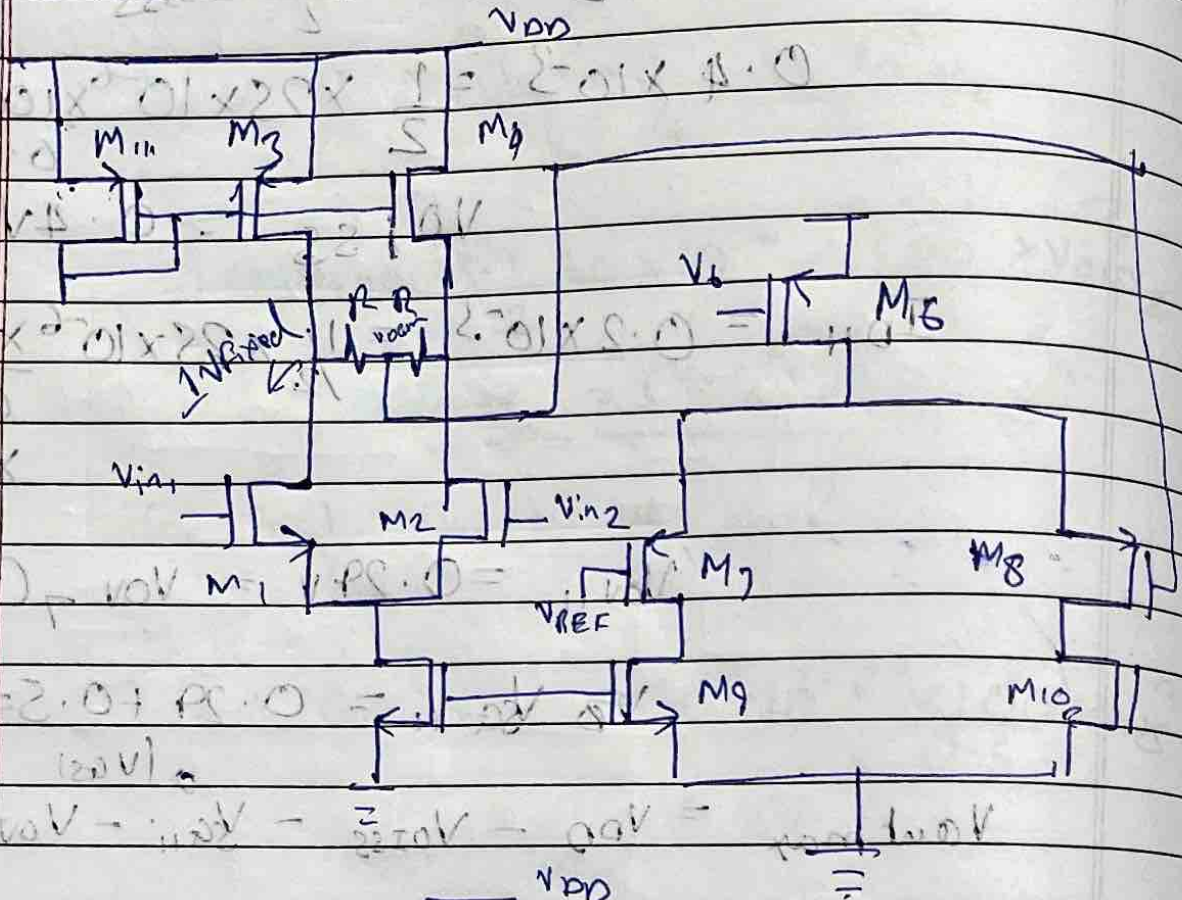
Now start 18/9/24 2nd loc

20/09/24

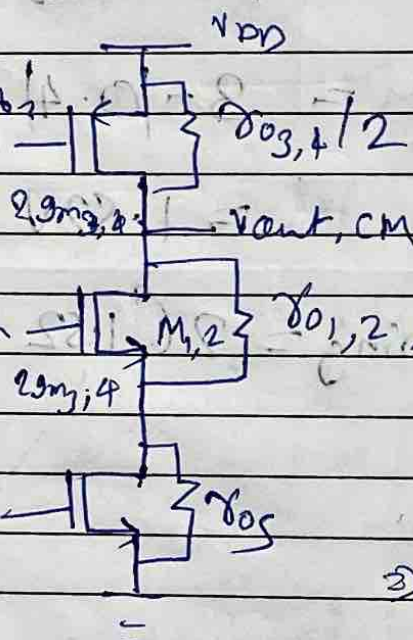
classmate

Date
Page

Common mode ckt



half ckt



$$\Rightarrow \frac{g_m r_o R_D}{R_D + R_S + r_o + g_m R_E}$$

$$V_{out\ min} = V_{DD}$$

→ 0.4

$$V_{out\ max} = V_{DD} - V_{DS,sat} \rightarrow 1.6$$

avg 1.1V

$$\Rightarrow \frac{2g_{m1,2} \times \frac{r_{o1,2}}{2} \times \frac{r_{o3,4}}{2}}$$

$$\frac{r_{o3,4}}{2} + r_{o5} + \frac{r_{o1,2}}{2} + 2g_{m1,2} r_{o5} \times \frac{r_{o1,2}}{2}$$

 $\approx$

$$\frac{r_{o3,4}}{2} + r_{o5} + \frac{r_{o1,2}}{2} + 2g_{m1,2} r_{o5} \times \frac{r_{o1,2}}{2}$$

$$\Rightarrow \frac{g_{m1,2} r_{o1,2} r_{o3,4}}{r_{o3,4} + 2g_{m1,2} r_{o5}}$$

$$\text{Gain} \Rightarrow \boxed{\frac{r_{o3,4}}{2r_{o5}}}$$

$$V_{in cm} = 1 + V_{th} = 1.5V$$

$$V_{in cm max} = 0.9V$$

$$\text{We are allowed to vary: } 1 \pm \frac{1}{100} \times 1$$

$$\downarrow 0.01$$

$$\downarrow 1V$$

$$\downarrow 0.01$$

$$0.99$$

$\Rightarrow$ ideal common mode gain

$\Delta V_{ocm \max}$ $\times$ Common mode voltage $\Rightarrow A_{ocm}$

$\Delta V_{in \text{ cm max}}$

$$A_{of} = \frac{A_{ocm}}{1 + \beta A_{ocm}}$$

~~$R_{out} \rightarrow r_{o3,4}$~~

~~$\frac{r_{o3,4}}{2}$~~

~~$r_{o5} + \frac{1}{2g_{m3,2}}$~~

or $r_{o5} + \frac{1}{2g_{m3,2}}$

$$\frac{r_{o3,4}}{2} \approx \frac{r_{o3,4}}{2r_{o5} + \frac{1}{g_{m3,2}}} \quad \text{neglect}$$

$$A_{ocm} = \frac{r_{o3,4}}{2r_{o5}} \approx \frac{1}{2}$$

$$A_{of} = \frac{r_{o3,4}}{2r_{o5}}$$

$$1 + \beta \frac{r_{o3,4}}{2r_{o5}} \ll 1$$

don't neglect it

$$\Rightarrow 0.5 < 0.03 (1 + 0.5\beta)$$

$$\Rightarrow 0.5 < 0.03 (1 + 0.5 \times 0.03 \times \beta)$$

$$-31.33 < \beta$$

$$A_f = \beta \approx g_{m8} \left(r_{o8} \parallel r_{o9} \parallel \frac{1}{g_{m9}} \right)$$

$$= \frac{g_{m8}}{2g_{m9}}$$

$$\hookrightarrow > 31.33$$

$$\beta \approx g_{m8} = \sqrt{2\mu_p C_{ox} \left(\frac{W}{L}\right)_8 I_{D8}}$$

$$\gg 31.3$$

$$2 \sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right)_9 I_{D9}}$$

$$\beta \approx \left(\frac{W}{L}\right)_8$$

$$> K$$

$$\left(\frac{W}{L}\right)_9$$

$$(4.2) I_{D7} = I_{D9} = 0.0 > 2.0 \text{ known}$$

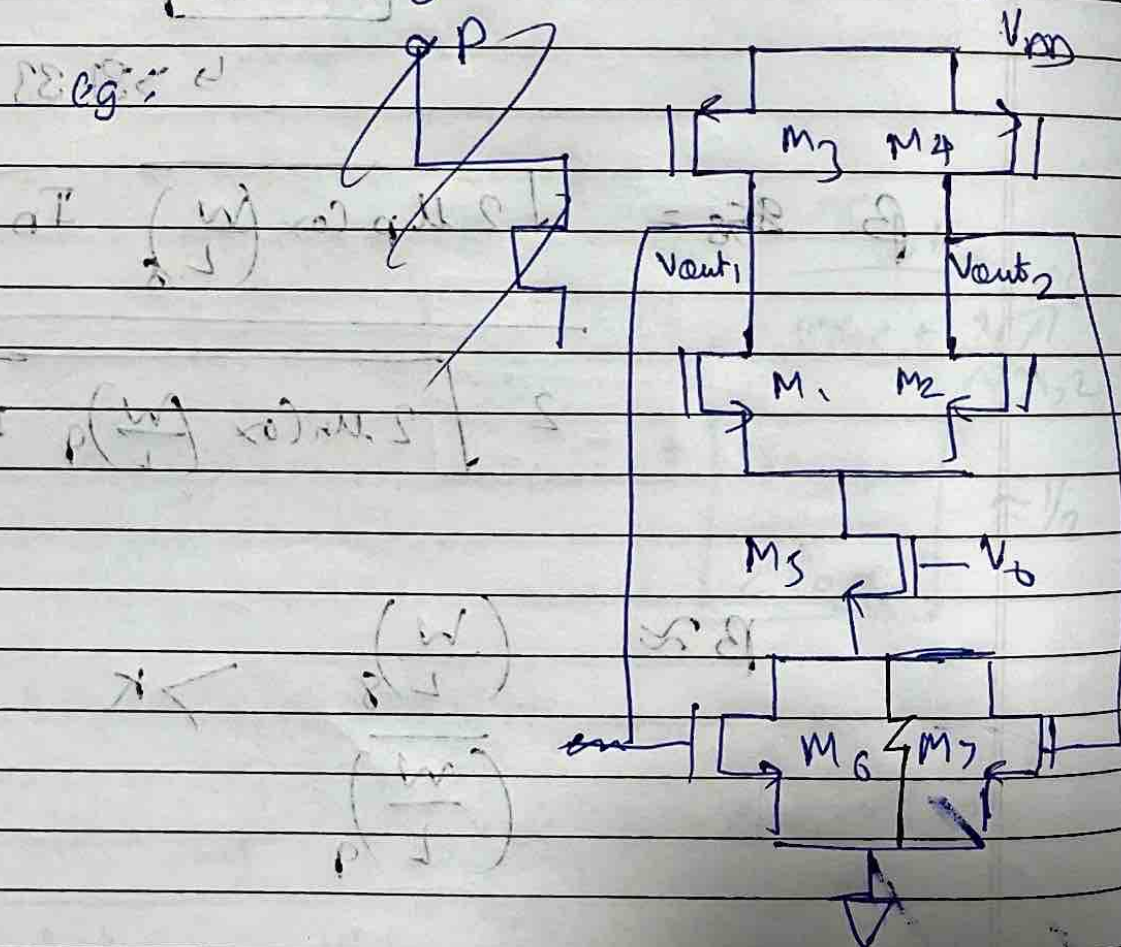
$$\Rightarrow \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_7 (V_{ov7})^2 = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_9 (V_{ov9})^2$$

$$8. > 18.18.$$

$$\Rightarrow \left(\frac{W}{L} \right)_7 = \frac{\mu_n C_{ox}}{\mu_p C_{ox}} \cdot \frac{(V_{ov9})^2}{(V_{ov7})^2}$$

$\Rightarrow R$ is very large here but size also $\uparrow$ creating problems

$\therefore$ Combining sensing ckt & our ckt into one



$$I_D = \mu_n C_{ox} \frac{W}{L} \left[V_{GS} - V_{th} - \frac{V_{DS}}{2} \right] V_{DS}$$

↳ in deep triode region

↑
Smaller
factor

$$\therefore I_D = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th}) V_{DS}$$

↳ eqn of an linear resistor

$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})}$$

$$R_{on1}/R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{out1} - V_{th})}$$

$$|| \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{out2} - V_{th})}$$

⇒

$$\frac{1}{\mu_n C_{ox} \frac{W}{L} [V_{out1} + V_{out2} - 2V_{th}]}$$

If black res is added (Better)